

Ministry of High Education

Culture and Science City on 6 October City

The High Institute of Computer Science & Information Systems



المعهد العالي لعلوم الحاسب و نظم المعلومات

Graduation Project:

**A Comprehensive Platform for Monitoring Infant Behavior
and Maternal Guidance .**

Supervised by:

Dr. Shimaa Mahmoud Dr. Ebtesam El-Hossiny

Project No: 606

Academic Year: 2024/2025

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(Sorted by ID number & in Arabic)

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Project Abstract

This project presents a comprehensive digital platform designed to support new mothers in caring for their infants by providing a range of tools and features that simplify the motherhood experience. The platform is available as both a mobile application and a website, offering mothers the flexibility to access its services on their preferred devices.

The platform enables users to create personal accounts, allowing them to connect with other mothers, exchange experiences, and share advice, fostering a strong sense of community. Additionally, mothers can upload images and list items their children no longer need, such as toys, strollers, and clothing, to sell directly to other mothers without intermediaries. This facilitates easy reuse and offers cost-effective solutions for families.

One of the standout features of the platform is an AI-powered voice recognition system, which allows mothers to record their babies' cries for analysis. The AI then identifies potential reasons for the crying, such as hunger, colic, or the need for sleep, helping mothers better understand and respond to their infants' needs more quickly and effectively.

Furthermore, the platform offers a comprehensive educational resource center with articles and expert guidance on infant care, equipping mothers with essential knowledge and skills to support their child's health and development. It also provides valuable information on common illnesses that infants may experience at different developmental stages, helping mothers stay informed and take preventive measures when necessary.

By integrating artificial intelligence with community-driven support and educational content, this platform aims to enhance the motherhood experience by delivering innovative solutions that assist mothers in their journey. Ultimately, it seeks to improve the well-being of both mothers and their children, making it a valuable tool for any mother looking for comprehensive and effective support.

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Chapter 1: Introduction

1.1 Overview

Motherhood is a life-changing experience, yet it comes with numerous challenges , particularly for first-time mothers [1]. Understanding an infant's needs, interpreting cries, and managing baby care efficiently can be overwhelming. In recent years, technological advancements, particularly in Artificial Intelligence (AI), have provided new solutions to assist parents in childcare.

With the increasing reliance on AI-driven applications in healthcare and personal assistance, leveraging AI in motherhood support has become an innovative approach [2]. This project introduces an AI-powered platform designed to support new mothers by integrating **voice recognition, a marketplace for baby products, and a community-driven support system** to enhance the parenting experience.

This platform enables mothers to **analyze their baby's cries using AI**, facilitating better understanding of their needs. Additionally, the **Marketplace feature** allows them to exchange baby products conveniently, reducing costs and promoting sustainability. The **community feature** fosters engagement among mothers, offering them a space to share experiences and seek guidance from one another [1].

1.2 Features of the Platform

This platform incorporates three major components designed to address the key challenges faced by new mothers:

- **AI-Powered Cry Analysis:** Allows mothers to record or upload their baby's cry, which is analyzed to identify potential causes such as hunger, discomfort, or sleepiness [2].
- **Marketplace for Baby Products:** Enables mothers to buy and sell second-hand baby products efficiently, reducing costs and promoting sustainability.
- **Community Chat:** supportive environment to share experiences, ask questions, and exchange parenting tips — fostering a sense of belonging and mutual support among moms [1].
- **Educational resources:** A digital guide that offers daily tips and trusted content to support mothers through various stages of child development, helping them provide better care with confidence.

1.3 AI-Powered Cry Analysis

Artificial Intelligence (AI) has revolutionized various fields, including healthcare and childcare. AI-powered sound recognition systems can analyze different types of sounds to extract meaningful insights. In the context of parenting, AI can assist in identifying the needs of infants by analyzing their cries and providing parents with potential explanations and recommended actions.

Newborns communicate primarily through crying, which can indicate hunger, discomfort, tiredness, or other needs. Traditionally, parents rely on experience and trial-and-error methods to interpret their baby's cries.

However, AI-powered systems can offer **data-driven insights** to help parents make informed decisions quickly.

In our platform, AI-powered cry analysis allows mothers to **record or upload their baby's cry**, which is analyzed to detect possible reasons for distress such as hunger, discomfort, or tiredness. The system then provides relevant recommendations to help mothers respond effectively. This feature helps alleviate parental anxiety and enhances their ability to cater to their child's needs more efficiently.

1.4 Marketplace for Baby Products

Online marketplaces have become a crucial part of modern commerce, enabling users to buy and sell products conveniently. Many parents struggle with baby items that are no longer needed but are still in good condition. A dedicated marketplace can help parents sell and purchase second-hand baby products at affordable prices, promoting sustainability and cost savings.

Parenting often involves purchasing many baby-related products, some of which are only useful for a limited time. Many parents find it difficult to resell these items or purchase necessary products at reasonable prices. Online platforms help bridge this gap by providing a safe and efficient environment for exchanging baby products.

Our platform integrates a **marketplace feature**, allowing mothers to **list and sell** baby-related products directly to other mothers. Buyers can browse available items, filter by category, and contact sellers through direct messaging or phone calls. This feature ensures that baby products find new homes instead of being discarded, reducing waste and benefiting families financially.

1.5 Community Chat

The **Community Chat** is a real-time group conversation space where mothers can communicate freely with one another. It allows them to share personal experiences, ask questions, and offer emotional support. This feature is designed to foster a sense of community, helping mothers feel connected and less isolated during the early stages of parenthood. The chat is open to all users, creating a welcoming environment where peer interaction plays a central role in knowledge-sharing and emotional relief.

1.6 Educational Guidance

The **Educational Guidance** section of the platform is designed to provide mothers with structured, accessible, and expert-approved information related to baby care and parenting. It contains ready-made, categorized content that mothers can explore at any time without needing to interact with others. The tips offered in this section cover a wide range of essential topics, including baby nutrition, sleep routines, hygiene practices, emotional development, and overall well-being.

This guidance is curated to ensure clarity, accuracy, and relevance for different stages of early motherhood. Whether a mother is looking for help with feeding schedules or understanding her baby's emotional needs, this section acts as a reliable and easy-to-navigate source of support. It empowers users to make informed decisions and build confidence in their daily parenting responsibilities.

1.7 Technologies Used

The platform is built using modern technologies to ensure efficiency, scalability, and security:

- Frontend: React.js (Web) and Flutter (Mobile)
- Backend: .net , flask
- Database: SQL server
- AI Model [3]: TensorFlow for voice recognition
- server : Monster API

1.8 Project Scope

The AI-powered motherhood support platform is designed to provide a comprehensive solution for new mothers by integrating multiple essential services. The project scope includes:

- **AI Voice Recognition System:** This feature allows mothers to upload or record their baby's cry, which is then analyzed to provide potential causes such as hunger, sleepiness, or discomfort.
- **Marketplace:** A peer-to-peer buying and selling platform where mothers can list, purchase, and exchange second-hand baby products.
- **Community Support & Chat System:** A real-time chat feature that enables mothers to communicate, seek advice, and share experiences in group conversations.
- **Educational Guidance & Parenting Tips:** A section within the application that provides essential baby care tips and expert advice. These structured guidelines help mothers understand different aspects of childcare, covering topics like feeding, sleep routines, and developmental milestones.
- **Cross-Platform Availability:** The platform will be accessible via a mobile application and a web-based interface to ensure usability across different devices.
- **Scalability & Security:** The system is built to support an increasing number of users while maintaining data privacy and security through encryption and authentication mechanisms.

The project primarily targets new mothers who seek AI-assisted parenting tools, cost-effective baby products, and reliable parenting guidance. Future expansions may include additional AI-based features, multilingual support, and integration with healthcare services.

1.9 Organization

- Chapter 1: Introduction: This chapter introduces the project, its objectives, and the problems it aims to solve. It also provides an overview of the technologies used in the system.
- Chapter 2: Covers the theoretical background and technologies used in the project , such as AI, databases, cloud computing, and frontend/backend frameworks.
- Chapter 3: System Analysis & Design: This chapter explains the system architecture, data flow, and design decisions, including UML diagrams that illustrate system interactions and database structure.
- Chapter 4: Explains the implementation phase: This section outlines the development process, including frontend and backend implementation, AI-based cry detection, database management, real-time communication, and security measures.

1.10 Conclusion

This chapter explored the theoretical background and technologies that support the AI-Powered Motherhood Support Platform. We discussed how artificial intelligence, machine learning, and voice recognition contribute to baby cry analysis. Additionally, we covered the significance of online marketplaces for baby products and community-driven support systems in enhancing the motherhood experience. These foundations set the stage for the following chapters, where we will dive deeper into system analysis, design, and implementation.

Chapter 2: Theoretical background and tools

This chapter provides a comprehensive overview of the theoretical concepts and technological tools that form the foundation of the proposed system. Each component of the system—Marketplace, AI-based Voice Recognition, Chat System, and Mother Guides—relies on specific theories and technologies that ensure functionality, usability, and scalability. By understanding the underlying principles and tools, we can better appreciate the structure and design decisions made throughout the development of the application. This chapter explores the essential background and the key technologies employed in building each module of the system.

2.1 User Authentication and Email Services

In modern software applications, authentication systems and email services are essential for ensuring security, personalizing user experiences, and facilitating effective communication between the system and users. These systems are widely used in platforms such as e-commerce websites, social media applications, educational systems, and healthcare services **as shown in Figure 2.1** [4].

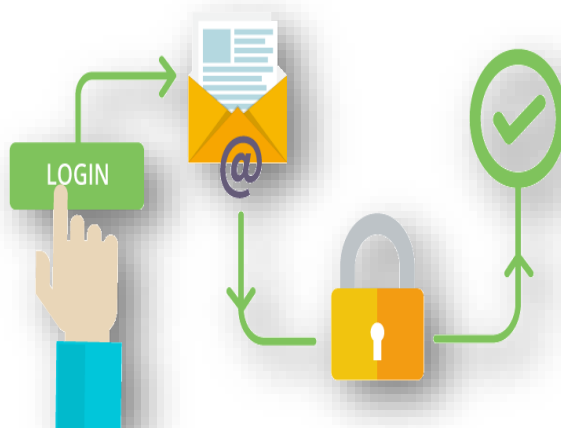


Figure 2.1 User authentication and email verification flow

Types of Authentication Systems

- **Password-Based Authentication:** A traditional method where the user enters an email or username along with a password.
- **Multi-Factor Authentication (MFA):** Adds an extra layer of security by requiring two or more factors (such as a password and a code sent to the phone or email).
- **Biometric Authentication:** Relies on unique biological traits like fingerprints, facial recognition, or voice.
- **Token-Based Authentication:** Common in web applications, where a temporary token is generated for the user to access protected resources.
- **OAuth & Social Login:** Allows users to log in using their accounts from external services like Google or Facebook, without creating a new account.

Email Services in Software Systems

Email services are crucial for performing various tasks, such as:

- **Account Verification:** By sending a verification email during registration.
- **Password Recovery:** Allowing users to reset their password securely.
- **Two-Factor Authentication (2FA):** Sending one-time codes via email for additional verification.

Common tools and protocols used:

- **SMTP:** Email sending protocol.
- **Gmail API / Outlook API:** For integrating with popular email services.
- **SendGrid / Mailgun:** Third-party services for reliable and secure email delivery.

Authentication and Password Recovery in the System:

A secure login and password recovery system was developed using Token-Based Authentication and Email Verification. Users can create accounts and sign in to access features like baby cry analysis, AI chat, and a marketplace.

In case of forgotten passwords, the user submits their email to receive a one-time verification code via Gmail. Once verified, the user can reset their password securely.

Password Recovery Flow :

- User enters their email on the "Forgot Password" page.
- System checks if the email exists.
- A temporary verification code is generated.
- The code is sent to the user's email.
- User enters the code to verify identity.
- A new password can then be set.

Tools and Technologies Used:

- ASP.NET Core Identity – Authentication and user management.
- JWT (JSON Web Tokens) – Token generation for secure access.
- SQL Server – Storing user data and verification codes.
- Gmail SMTP – Sending verification codes.
- React & Flutter – Building web and mobile interfaces for login and recovery.

2.2 Educational Guidance Systems

Educational guidance systems are digital tools designed to deliver relevant and helpful knowledge to users in a structured, user-friendly manner. These tools are widely implemented in applications that aim to inform, guide, or support users—particularly in fields like healthcare, education, and parenting.

Unlike dynamic interactions such as chats or forums, educational guidance provides **static or pre-curated content** that is carefully selected to offer value without requiring user input. It ensures accessibility to accurate and reliable information at any time.

These systems can take the form of tooltips, tip-of-the-day features, tutorials, or categorized advice panels. Their goal is to simplify access to knowledge, support decision-making, and reduce dependency on external sources or real-time support.

In our platform, we included a dedicated **Guidance Module** that provides new mothers with parenting advice and practical tips. The module displays **10 randomly selected tips** each time the user accesses the section, covering key areas such as:

- Baby care
- Nutrition
- Hygiene
- Sleep routines

This randomness helps maintain user engagement and ensures varied content exposure over time.

We designed this module using simple, readable cards grouped under relevant themes. Each card provides short, actionable tips without overwhelming the user. The tips are fixed and system-generated, requiring no search or interaction from the user.

This educational guidance was implemented using frontend components (accordion-style layout), powered by lightweight backend logic to handle the randomization. It acts as a built-in resource center to help mothers confidently navigate daily parenting tasks.

2.3 Online Marketplace For Baby Products

This section discusses the concept of the Marketplace in modern applications, where it serves as an intermediary platform that allows users to display and sell their products easily and securely. Marketplaces are considered one of the essential components of digital commerce applications, offering an interactive environment between sellers and buyers without the need for a third party. Some well-known global examples include Amazon, eBay, and Etsy, which may differ in their business models but share the common goal of facilitating online buying and selling **as shown in figure 2.2.**



Figure 2.2 Online Marketplace for Baby Products

Some of the key theoretical aspects in designing and building marketplace systems include [6]:

- **E-commerce Models:** One of the most prominent models is C2C (Consumer to Consumer), where users can directly list and sell their products to other users — which is the model used in the proposed system.
- **User Experience (UX):** A well-designed, user-friendly interface is essential for the success of any marketplace. This includes organizing product listings, the ability to save items, and easy communication with sellers.
- **Database Structure:** A solid database is required to manage and store product information, user data, and relationships such as saved items or user-listed products.
- **Security Aspects:** Ensuring the protection of user data and preventing unauthorized access or manipulation is crucial.

In the proposed system, the marketplace has been designed specifically to meet the needs of mothers looking to buy and sell baby-related products. It offers a variety of features and functionalities, including:

- **Add Product:** Any user can upload a product through a form that includes the product name, description, price, images, category, and a phone number — allowing direct communication between buyer and seller without any intermediary.
- **Product Details Page:** Each product has a dedicated page showing full details including images, description, price, and the seller's contact number.
- **Save Product:** Users can save products to a “Saved Products” list by clicking the heart icon displayed below each product image.
- **Search and Filtering:** The system provides keyword-based search functionality along with filters by category or price to help users easily find the desired products.
- **Delete Product:** Users can delete the products they’ve uploaded from the “My Products” page, giving them full control over their listings.

This marketplace model is designed with simplicity and practicality in mind, encouraging direct community-based interaction among mothers without involving complex payment gateways or third-party mediators.

Tools Used:

- React: For developing the web-based marketplace interface.
- Flutter: For building the mobile application interface.
- SQL Server: For storing data related to products, users, and saved items.

2.4 Community Chat

Chat systems have become an essential part of modern mobile and web applications, allowing users to communicate, share information, and engage in real-time or asynchronous conversations. From social platforms like WhatsApp, Telegram, and Facebook Messenger, to chat components embedded in e-commerce, education, and healthcare apps — chat functionality has proven to significantly enhance user engagement and satisfaction **as show in figure 2.3.**

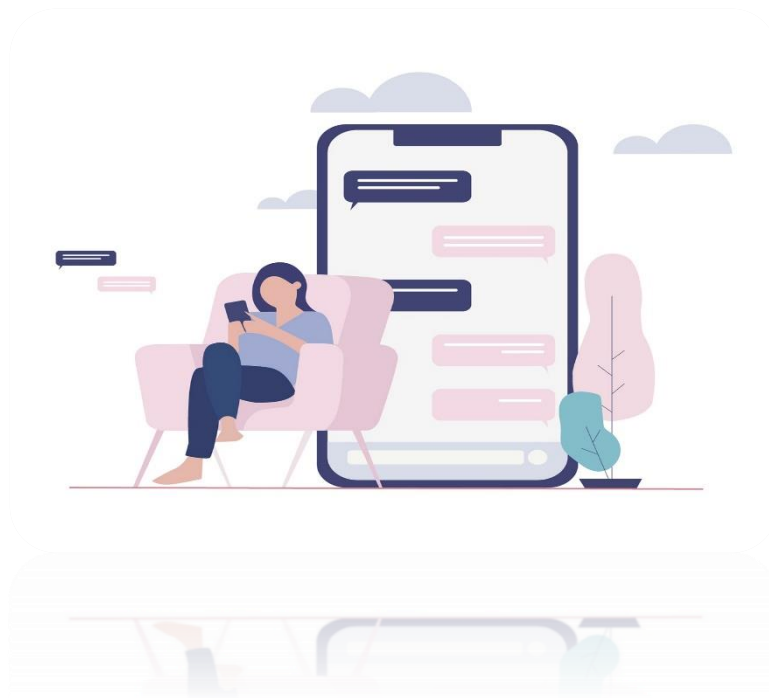


Figure 2.3 Community Chat

There are various types of chat systems, including:

- Private Messaging (one-to-one communication),
- Group Chats (multi-user communication within shared spaces),
- Admin-controlled Chats (where certain users have moderation rights),
- Anonymous or Public Chats (open to all users without restrictions).

In the proposed system, a simple group chat functionality is implemented to support communication among all users of the app — specifically mothers — allowing them to interact, share advice, experiences, and support each other. The chat design follows a straightforward and inclusive approach, with the following characteristics:

- Public Group Chat Only: All users who sign up in the application have access to a single shared group chat. There are no admins or private messaging features.
- Message Sending: Any user can send messages freely to the group.
- Message Editing: Users can edit the messages they've previously sent.
- Message Deletion: Users can delete a message for themselves (i.e., it will no longer appear in their chat view).
- Users also have the option to delete a message for everyone, in which case it will be removed from the group view for all users.

2.5 Artificial Intelligence (Voice Recognition)

Artificial intelligence (AI) has become an essential element in many modern applications, used to enhance user experience, automate tasks, and make smart decisions based on data. Some of the prominent fields of AI that are widely used today include:

- Image Recognition: Identifying images and objects within them.
- Natural Language Processing (NLP): Understanding and analyzing human language.
- Recommendation Systems: Recommending products or content.
- Speech and Voice Recognition: Analyzing voice and recognizing words, emotions, or states.

In the proposed project, voice recognition technology is used to analyze a baby's cry and attempt to determine the possible cause of the crying, whether it is hunger, pain, sleepiness, or other reasons. The goal of this feature is to help mothers understand their children's needs more quickly and easily through scientifically-backed AI analysis **as shown in figure 2.4.**



Figure 2.4 Artificial Intelligence (Voice Recognition)

Types of Voice Recognition Technologies:

- **Speaker-Independent Recognition:** This type of voice recognition does not depend on the speaker, as it can recognize voice regardless of who the speaker is. It relies on training the model on a wide range of voices from different individuals. In the context of the project, this can help identify the cry of any baby, no matter who they are.
- **Speaker-Dependent Recognition:** This type of recognition relies on the specific voice of a particular speaker. The model needs special training for that person and is often used in applications like personal voice assistants. In the project, this type could be used to analyze the voice of a particular child based on special training.
- **Continuous Speech Recognition:** This type allows for recognizing words continuously without needing to pause between sounds. It can be used in applications that require ongoing analysis of crying or spoken commands.
- **Emotion Recognition in Voice:** This type involves recognizing emotions from the voice. By analyzing the tone of voice, the system can determine whether the voice expresses emotions like anger, joy, sadness, or anxiety. In your project, this type could help analyze a child's emotions based on the tone of the cry, which could then assist in determining the likely cause of the cry.

Features of the AI System in the Project:

- Upload Audio or Live Recording: The user can either record the child's cry directly through the app or upload a ready audio file.
- Prediction of Cry Cause: The voice is analyzed using an AI model, and the user is given a prediction of the cause of the cry (such as "Hunger", "Pain", "Need to sleep", etc.).
- Random Interaction Phrases: The same sentence is not sent every time, even if the prediction is similar; a different sentence is chosen each time. For example, if the prediction is "Hunger", the user might see sentences like: "It seems like he's hungry, try breastfeeding him." , "Maybe it's time for food, try offering milk."
- Delete Recording or Prediction: Users can easily delete recordings or prediction results.

Why This Type of AI?

Voice Recognition was chosen because it is the most suitable for the use case, as a baby's cry is one of the primary ways they express their needs. By analyzing the pitch, duration, and pattern of the cry, the model can be trained to detect patterns and determine the likely cause.

Tools Used:

- Flask (Python): For creating a custom API for AI and voice analysis.
- TensorFlow / PyTorch (Deep Learning frameworks): To train the deep learning model for voice recognition and sound classification.
- Deep Learning Techniques: Used to build and train complex models that can detect patterns in baby cries with high accuracy.
- Librosa: For analyzing sound signals and converting them into Spectrograms used in training.
- ASP.NET: For linking the user interface to the smart services and handling data processing.
- SQL Server: For storing recordings, prediction results, and users.
- React: For developing the web interface for the AI feature.
- Flutter: For designing the mobile interface for uploading voice recordings and displaying predictions.

2.6 Technologies Used

Technology plays a crucial role in the development of modern applications, enhancing efficiency, scalability, and user experience. The AI-Powered Motherhood Support Platform integrates various technologies to ensure a seamless and intelligent experience for mothers. This section provides an overview of the key technological components utilized in the project.

2.6.1 Front Development

Front development is responsible for designing and building the user interface (UI) that users interact with. It focuses on creating a visually appealing, responsive, and user-friendly experience **as shown in figure 2.5.**



Figure 2.5 Technologies used in Front-end

Frontend technologies include:

- **HTML, CSS, and JavaScript:** The foundation of web development, ensuring structure, styling, and interactivity.
- **React:** is a UI library used for building interactive user interfaces, particularly for single-page applications. It emphasizes a component-based architecture, allowing developers to create reusable UI components. React utilizes a virtual DOM to optimize rendering and improve application performance.
- **Angular:** Angular is a comprehensive framework based on TypeScript, designed for building large-scale, single-page applications. It provides a structured approach with features like two-way data binding, dependency injection, and a modular architecture. Angular is suitable for complex applications requiring a robust solution.
- **Vue.js:** Vue.js is a progressive framework for building user interfaces. It is designed to be incrementally adoptable, focusing on the view layer only. Vue.js combines features from both React and Angular, offering simplicity and flexibility, making it suitable for both small and large-scale applications.
- **Bootstrap:** Bootstrap is a popular CSS framework that provides pre-designed components and responsive grid systems to facilitate quick and consistent web design. It focuses on the visual and structural aspects of web development, offering a collection of CSS and JavaScript tools for building responsive and mobile-first websites.
- **Flutter:** A UI toolkit developed by Google for building natively compiled mobile applications, ensuring smooth performance on both iOS and Android and it enables developers to create natively compiled applications for mobile, web, and desktop from a single codebase. It uses the Dart programming language and provides a rich set of pre-designed widgets, allowing for highly customizable and responsive designs

In this project: The frontend is designed to provide an intuitive experience for mothers, ensuring seamless navigation across both web and mobile platforms using **React for web development and Flutter for mobile applications** for this reasons :

- i. **React:** We chose React.js in our project due to its multiple advantages that align well with our goals. React's use of the virtual DOM enhances performance by minimizing costly updates, ensuring a smooth user experience. Its component-based architecture allows developers to create reusable and independent components, making the application more maintainable and scalable.

Additionally, React adopts one-way data binding, which improves data flow control and simplifies debugging, resulting in more predictable behavior. Another key factor is React's flexibility; as a library rather than a full-fledged framework, it easily integrates with various tools, libraries, and backend services, offering the adaptability needed for our project. Lastly, React boasts strong community support, providing a wealth of educational resources and third-party libraries that accelerate development and problem-solving.

- i. **Flutter:** it was selected for this project due to its multiple advantages that align with the goals of efficient and seamless mobile application development. It enables cross-platform development using a single codebase, significantly reducing development time and effort while ensuring consistency across platforms.

Flutter's high performance is achieved through its own rendering engine, providing smooth animations and excellent user experience. Additionally, Flutter benefits from strong support and backing by Google, resulting in a rapidly growing community, comprehensive documentation, and an abundance of libraries and tools that simplify the development process. Its control over every pixel ensures consistent UI design and behavior across different devices, minimizing issues related to platform-specific variations.

2.6.2 Backend

Backend development is a fundamental aspect of modern software applications, responsible for managing the server-side logic, processing requests, and handling database interactions. It serves as the backbone of any application, ensuring seamless communication between the frontend and the database while maintaining security, performance, and scalability **as shown in figure 2.6.**

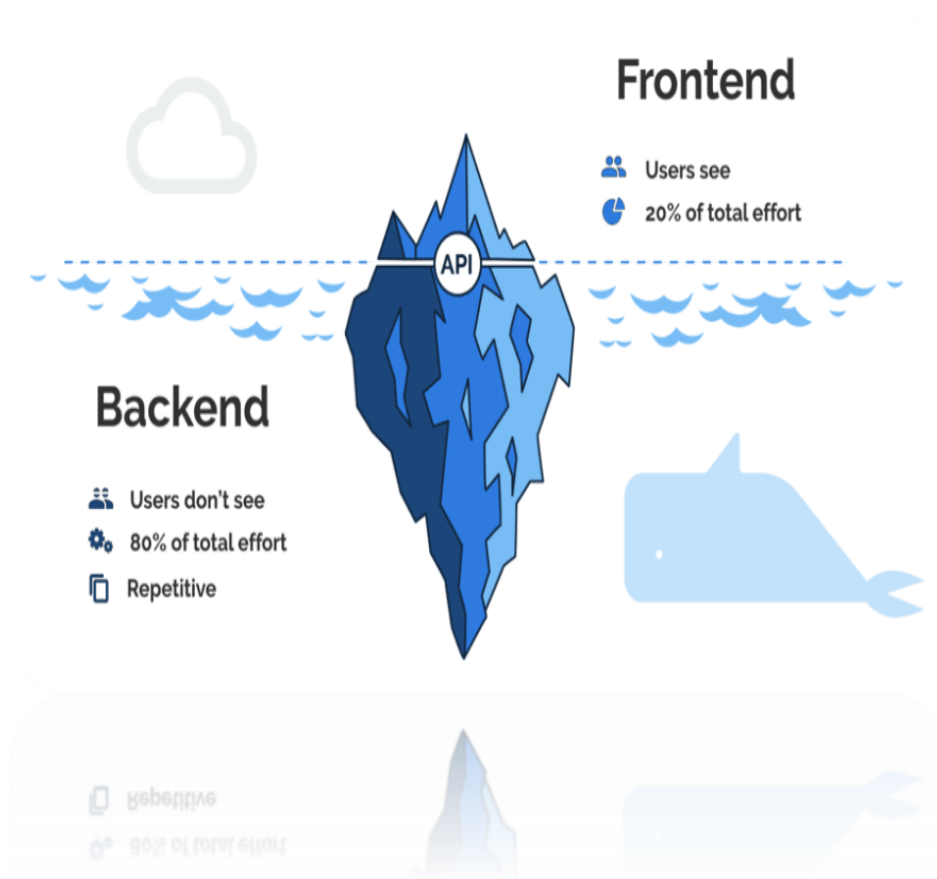


Figure 2.6 Technologies used in Back-end

Backend frameworks and technologies enable developers to build robust and efficient systems that can process user inputs, authenticate users, store and retrieve data, and integrate with external services include:

- **Django:** A high-level Python web framework that encourages rapid development and clean, pragmatic design. It includes numerous built-in features, such as an ORM, authentication, and an admin interface,
- **ASP.NET Core:** is an open-source, cross-platform framework used for building modern, cloud-based, and internet-connected applications. It is known for its high performance and supports multiple operating systems, including Windows, Linux, and macOS, making it a flexible choice for development.

The framework seamlessly integrates with the Microsoft ecosystem and various services, enabling developers to create reliable and scalable applications with ease. With its support for asynchronous programming, optimized resource management, and advanced security features, ASP.NET Core stands out as one of the most powerful frameworks for developing web applications and cloud-based services.

- **Node.js:** is an open-source runtime environment based on JavaScript, used for developing web applications and servers. It efficiently handles multiple requests simultaneously due to its non-blocking, event-driven architecture, making it ideal for applications like real-time chat, APIs, and streaming services. It also provides the npm library, which offers thousands of ready-to-use packages to simplify development.

- **Laravel:** is an open-source PHP framework used for building web applications easily. It follows the MVC pattern to organize code and offers powerful features like Eloquent ORM for database management, Blade Template Engine for designing views, and routing for handling web requests. It also includes Artisan CLI, a command-line tool that simplifies development tasks. With these features, Laravel is a popular choice for creating modern and flexible web applications.
- **Spring Boot:** is a framework used with Java for developing web applications and backend services. It simplifies the development process by reducing configuration requirements and providing built-in tools. It allows developers to create APIs and work with databases easily, making it widely used for enterprise applications and cloud services.
- **Ruby on Rails:** is a backend framework built with the Ruby programming language. It is used for developing web applications efficiently and follows the MVC (Model-View-Controller) pattern, which helps organize code by separating data, business logic, and the user interface.

In this project: The backend is responsible for handling user authentication, AI request processing, marketplace transactions, and chat functionalities include:

- **ASP.NET Core:** **ASP.NET Core:** was selected for our project due to its high performance, cross-platform capabilities, and strong security features [5]. It offers **asynchronous programming** and **native code compilation**, ensuring faster response times and efficient resource utilization. The framework's **cross-platform nature** allows deployment on Windows, Linux, and macOS, providing flexibility in development. Backed by **Microsoft**, it benefits from extensive documentation, continuous updates, and built-in security features like authentication and data protection, making it a reliable choice for secure application development.
- **Flask (Python):** A lightweight web framework used to host and deploy AI models for baby cry analysis. It acts as a backend service that processes incoming audio recordings, analyzes them using machine learning models, and returns results through an integration with the .NET API.

2.6.3 Database

A database is a structured collection of data that allows efficient storage, retrieval, and manipulation of information. In modern software systems, databases play a critical role in managing data consistency, supporting transactions, enforcing relationships, and enabling secure and scalable access to information **as shown in figure 2.7.**

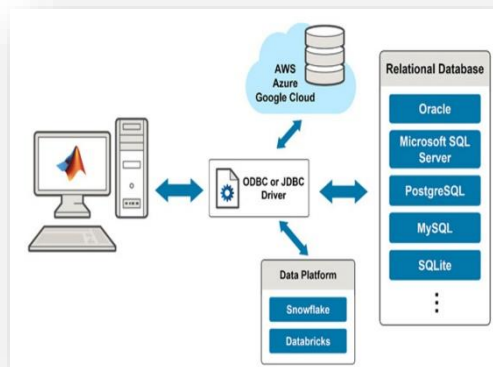


Figure 2.7 Technologies used in Database

There are two main types of databases:

- **Relational Databases (SQL):** Store data in structured tables with rows and columns. They use SQL (Structured Query Language) for querying and managing data. These databases support relationships between tables through keys and constraints
- **Non-Relational Databases (NoSQL):** Store data in flexible formats like JSON, key-value pairs, or graphs. They are schema-less and more suitable for unstructured or rapidly changing data.

Common Types of Databases

There are several database systems available in both SQL and NoSQL categories, the most popular ones include:

- Microsoft SQL Server: A relational database developed by Microsoft, known for its integration with the .NET ecosystem. It is ideal for enterprise-grade applications.
- MySQL: An open-source relational database widely used in web development, especially in PHP and LAMP stack projects.
- PostgreSQL: A powerful, open-source relational database that supports advanced data types and full SQL compliance.
- MongoDB: A leading NoSQL document-oriented database that stores data in JSON-like formats. It is highly scalable and flexible.
- Oracle Database: A high-performance commercial relational database system used by large enterprises for mission-critical applications.
- Firebase Realtime Database / Firestore: NoSQL cloud-hosted databases optimized for real-time data syncing, widely used in mobile apps.

In this project: we chose Microsoft SQL Server, a relational database management system (RDBMS), for the following reasons:

- Strong integration with ASP.NET Core: SQL Server works seamlessly with Entity Framework Core, allowing code-first database creation and efficient data handling.
- Support for complex relationships: Our system includes entities such as users, products, and categories, which are highly relational. SQL Server handles these relationships efficiently.
- High reliability and performance: SQL Server offers robust performance and data integrity, making it suitable for real-world scalable applications.
- Security features: It provides role-based access control, data encryption, and strong backup & recovery mechanisms.

2.6.4 AI Model

Artificial intelligence (AI) is one of the most transformative technologies in the modern era, enabling machines to perform tasks that traditionally require human intelligence. AI encompasses various subfields, including Machine Learning (ML), Deep Learning, Natural Language Processing (NLP), and Computer Vision. These technologies allow systems to analyze data, recognize patterns, and make informed decisions, improving efficiency across different domains **as shown in figure 2.8.**

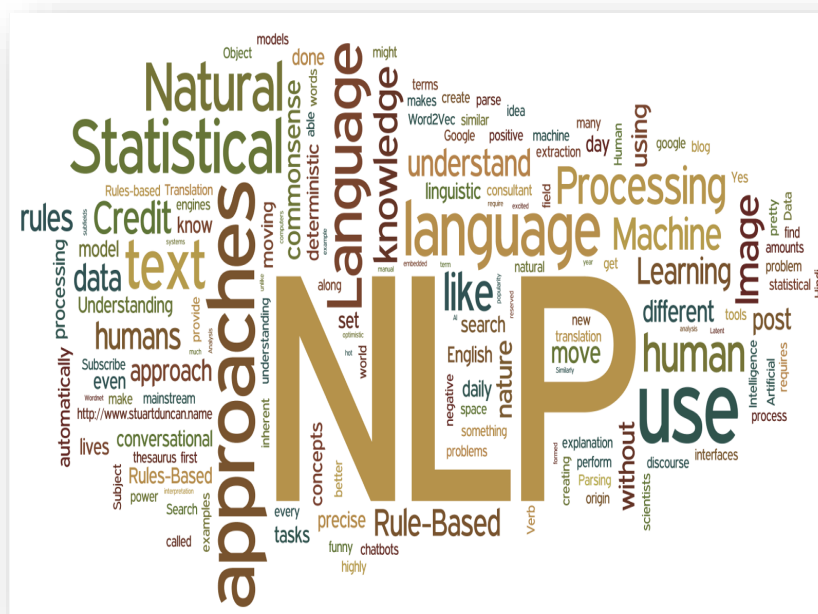


Figure 2.8 Technologies used in AI Model

AI can be categorized into three main types based on its capabilities:

- **Narrow AI (ANI - Artificial Narrow Intelligence):** Specialized in performing specific tasks, such as speech recognition and recommendation systems.
- **General AI (AGI - Artificial General Intelligence):** Aims to perform any intellectual task that a human can do, though it is still under research and development.
- **Super AI (ASI - Artificial Super Intelligence):** A theoretical concept where AI surpasses human intelligence in all fields, which has not yet been achieved.

Key AI technologies include:

- **Machine Learning (ML):** Enables systems to learn from data and improve performance over time.
- **Deep Learning:** A subset of ML that uses neural networks to process complex data and enhance decision-making.
- **Natural Language Processing (NLP):** Allows machines to understand and analyze human language, commonly seen in virtual assistants like Siri and Google Assistant.
- **Computer Vision:** Helps machines interpret and process visual data, widely used in medical imaging and security surveillance.

AI is applied in various fields such as healthcare (disease diagnosis), finance (fraud detection), cybersecurity (threat identification), and autonomous vehicles (self-driving cars). Despite its benefits, AI faces challenges like data bias, privacy concerns, and potential job displacement due to automation.

In this Project:

We implemented AI technology to enhance the functionality of the **Motherhood Support Platform**.

The AI-powered **cry analysis** system allows mothers to record or upload their baby's cry, which is analyzed using **machine learning models** to determine possible causes such as hunger, discomfort, or sleepiness. Based on the analysis, the system provides personalized recommendations to help mothers respond effectively to their child's needs. This feature aims to reduce parental anxiety and provide immediate, data-driven insights to improve baby care.

AI technologies enable applications to analyze data, recognize patterns, and provide intelligent recommendations. Machine learning and deep learning models help in developing AI-driven solutions tailored to user needs.

We used **Machine Learning (ML)** and **Deep Learning Models** to analyze baby cries, detect potential causes, and provide mothers with actionable insights. These models were trained on diverse datasets to ensure high accuracy and reliability in recognizing different baby cry patterns.

AI enhances the platform's ability to analyze and interpret data, offering personalized recommendations. Key AI technologies include:

- **Machine Learning (ML):** Enables the system to recognize and classify baby cries.
- **Natural Language Processing (NLP):** Helps improve chatbot responses and community interactions.
- **Deep Learning Models:** Used for advanced audio recognition and prediction

2.6.5 Servers

Server is a device or service responsible for hosting and running applications or websites to make them accessible to users over the internet. Servers are used in almost all modern systems—for storing data, handling user requests, or running entire applications. Choosing the right type of server depends on the size of the project, the required resources, and the level of control needed **as shown in figure 2.9.**

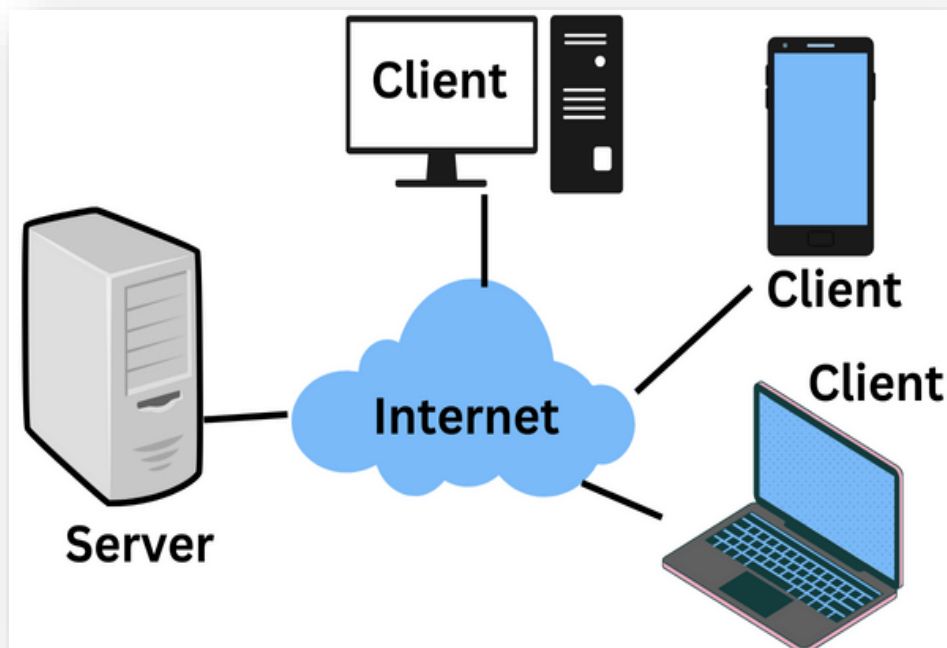


Figure 2.9 Technologies used in Server

Types of Servers:

There are several types of servers, each with different features and use cases:

- **Shared Hosting:** A simple and low-cost option where multiple websites share the same server. However, it offers limited performance and resources.
- **VPS (Virtual Private Server):** A virtual section of a larger server that provides more control and better performance than shared hosting. Suitable for medium-sized projects.
- **Dedicated Server:** A full server dedicated to a single project, offering high performance and flexibility, but it requires advanced management skills.
- **Cloud Server:** Offered by providers like AWS or Azure, cloud servers are scalable and flexible, but they require more complex configurations.
- **PaaS (Platform as a Service):** Platforms like Heroku or Monster API provide a ready-to-use environment for application deployment, without the need to manage server infrastructure manually.

In this Project:

We use Monster API because it is a PaaS solution, which is especially suitable for academic projects and prototypes, for several reasons:

It is easy to use and doesn't require complex configurations. It allowed me to deploy the backend of the entire application quickly and efficiently.

It is ideal for projects that need to be tested and demonstrated in a short time. It is cost-effective and doesn't require advanced knowledge of server management

2.7 Conclusion

In this chapter, we reviewed the theoretical foundations and technologies that form the core of the proposed system. Six main components were discussed: the marketplace, the AI-based voice recognition module, the group chat system, the mother guides section, the authentication system, and the email services. Key concepts such as user experience, machine learning techniques, communication models, data privacy, and user data handling were explored for each module.

In addition, the tools and frameworks used — including React, Flutter, ASP.NET, Flask, and SQL Server — were explained in terms of their roles in implementing each component. Token-based authentication and email verification methods (used for password reset and user validation) were also highlighted. This theoretical background provides a strong foundation for the system design and implementation phases, which will be discussed in.

Chapter 3: Analysis & Design

This chapter provides an analytical view of the proposed system, which is designed to support mothers by identifying the reasons behind a baby's crying using AI voice recognition. The system also includes a marketplace for baby-related products, a public group chat for communication between users, mother guides, as well as user authentication and email services.

System analysis is conducted to study the system or its components in order to identify its objectives. It allows us to focus on actual project development rather than spending time overthinking, as the modules are clearly defined. This ensures accurate estimation and successful implementation. In addition, data analysis helps in building a conceptual data model that defines both the structure of the data and the processes that operate on it.

3.1 Functional Requirement

These requirements mainly Specify, what the system has to do? It describes a software system or its component. A function is nothing but inputs to the software system, its behavior, and outputs. It can be a calculation, data manipulation, user interaction, or any other specific functionality which defines what function a system is likely to perform.

3.1.1 Project Function

In this section, we describe the essential functions and core features of the platform developed to support new mothers in caring for their infants. The system includes several modules designed to meet the needs of users across different tasks.

3.1.2 User Authentication

- The system allows users (mothers) to sign up and log in securely.
- The system must validate user credentials before granting access to ensure security and prevent unauthorized access.
- Email verification is required for new accounts, and users must have the ability to recover their passwords through an email-based reset option.
- The system shall prevent registration if the email is already used.
- The system shall store the user's credentials securely in the database.

3.1.3 User Management

- Users (mothers) can create accounts and log in securely.
- The system shall allow users to log in using valid credentials.
- Profile customization (name, email, phone number).
- Password recovery and authentication via email or phone verification.
- The system shall send a verification code to the user's email for password reset.
- The system shall validate all required fields during registration and login.

3.1.4 AI Powered Cry Analysis

- Mothers can **record or upload** a baby's cry for AI analysis.
- The AI system **analyzes the audio** and identifies possible reasons for distress (hunger, discomfort, tiredness, etc.).
- The system provides **recommendations** based on the analysis results.

3.1.5 Marketplace Functionality

- Users can **list** baby-related products for sale.
- Buyers can browse, filter, and search for products.
- Direct communication between buyers and sellers via **phone call or messaging apps (e.g., WhatsApp)**.
- The system shall allow users to save products by clicking a heart **icon**.
- The system shall allow users to view detailed information of a selected product.

3.1.6 Educational Guidance & Parenting Tips

- The system shall display 10 random mother guides on the home page.
- Step-by-step guides on baby care, health tips, and developmental milestones.

3.1.7 Community Chat

- The system shall allow mothers to participate in a community group chat without manual joining.
- The system shall allow sending and receiving text messages in real time using endpoints.
- The system shall allow users to see all messages sent in the group chat.
- The system shall display the sender's username and message timestamp.
- Delete Message for Everyone ; The system allows mothers to delete a sent message from the community chat for all users.
- Delete Message for Me ; The system allows mothers to delete a message from their own chat view without affecting other users' views.
- The system shall allow users to edit their own messages in the group chat.

3.2 Non Functionality Requirement

The non-functional requirements ensure the platform's efficiency, security, and scalability. Key non-functional requirements include:

3.2.1 Performance & Scalability

- The system must handle multiple concurrent users without performance degradation.
- AI processing should return cry analysis results **within a few seconds**.

3.2.2 Security & Privacy

- **End-to-end encryption** for user communication and stored data.
- Secure authentication using **OAuth or Firebase Authentication**.
- Compliance with data protection laws.

3.2.3 Usability & Accessibility

- Intuitive and easy-to-navigate user interface.
- Mobile-friendly design ensuring compatibility across different screen sizes.

3.2.4 Reliability & Availability

- System uptime should be **at least 99.9%** to ensure continuous availability.
- Automatic backups for preventing data loss.

3.3 Requirement Analysis

In this section, we use the modeling system “UML diagrams”, based on requirements analysis, to describe the functions and processes that the system must perform.

Introduction to UML:

Unified Modeling Language "UML" is a standardized graphical display format for the visualization, specification, design, and documentation of (software) systems. It offers a set of standardized diagram types with which complex data, processes, and systems can easily be arranged in a clear, intuitive manner.

UML Diagrams:

In this section, we present the UML diagrams that illustrate the dynamic and functional aspects of the system.

UML (Unified Modeling Language) diagrams are used to visualize, specify, construct, and document the structure and behavior of the system. These diagrams help in understanding the system's requirements and how different components interact with each other.

The diagrams used in this project include:

- Context Diagram
- Use Case Diagram
- Activity Diagram
- Sequence Diagram
- ERD Diagram
- Class Diagram

3.3.1 Context Diagram

A context diagram is a high-level diagram (Level 0) that shows the system's boundaries and how it interacts with external entities. It highlights the main data flows in and out of the system without detailing internal processes.

This model illustrates the overall interaction between the Motherhood System and the external entities that communicate with it. The diagram represents a high-level view (Level 0) of the system, focusing on the main data flows and how the system exchanges information with its surroundings without delving into internal processes **as shown in the context diagram.**

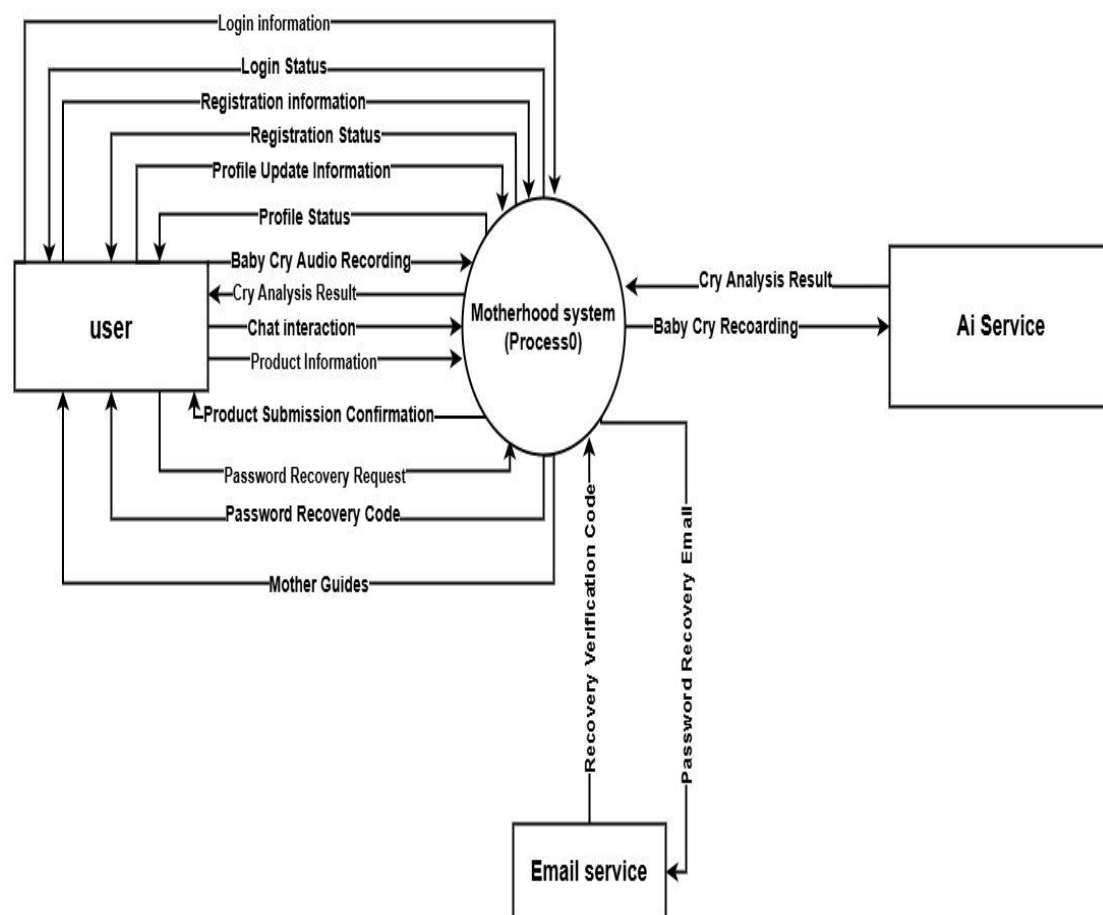


Figure 3.1 Context Diagram for Motherhood System

The system interacts with three primary external entities:

- **User:** The user is the primary actor, engaging in actions like registration, login, profile updates, submitting baby cry recordings for AI analysis, and managing marketplace activities (e.g., adding and browsing products). The system provides responses such as status updates, analysis results, educational guides, and notifications.
- **AI Service:** This service analyzes the baby cry recordings submitted by users. The AI model processes the recordings to predict the reason behind the baby's cry (e.g., hunger, discomfort, sleepiness), helping users understand and care for their babies.
- **Email Service:** The email service handles account security features, particularly password recovery. When a user requests a reset, the system sends a request to the email service, which sends a verification code or reset link to the user's email.

This context diagram provides a clear and concise overview of how the Motherhood System operates at the boundary level, emphasizing the essential communication and data exchange between the system and its external entities.

3.3.2 Motherhood system use case diagram

The Use Case Diagram describes the main functionalities provided by the system and how different users (actors) interact with these features. It helps to identify the system's scope and external interactions.

In this project, the main actor is the mother, who interacts with the application to perform various tasks. These include:

- Register and log in
- Record baby cries
- Browse and post in the marketplace
- Access guidance tips
- Communicate in group chats

The Use Case Diagram visually represents these interactions to clarify how the user engages with the core features of the system.



Figure 3.2 Main use case for Motherhood System

Use Case Table for Motherhood System:

Table 3.1 shows the main use case for the system.

Actors	Action	Module	Description
Mother	Register account (email, password)	User Authentication	Secure sign-up with email verification.
Mother	Login with credentials	User Authentication	Validates credentials for access.
Mother	Update profile (name, email, etc.)	User Management	Customizes personal details.
Mother	Upload/record baby's cry	AI Cry Analysis	Submits audio for AI processing.
Mother	View AI analysis results (hunger, discomfort, etc.)	AI Cry Analysis	Receives AI-generated insights and recommendations.
Mother	List product in marketplace (image, price, description)	Marketplace	Sells baby-related items.
Mother	Search/filter products	Marketplace	Browse or buy products.
Mother	Add/remove product from wish list	Marketplace	Saves favorite items.
Mother	Chat with other mothers (text, edit, delete)	Community Chat	Real-time group messaging with message control.
Mother	Access parenting guides (health tips, milestones)	Educational Guidance	Educational Guidance.
Mother	Reset password via email	Email Service	Recover account access securely.
System	Analyze cry audio (AI model)	AI System	Processes audio to identify baby's needs.
System	Send verification/password reset emails	Email Service	Integrates with SMTP/API
System	Store/retrieve messages, products, user data	Database	Manages data persistence

3.3.3 Activity Diagrams

The Activity Diagram illustrates the dynamic flow of activities within the system. It represents the sequential steps users or the system perform to accomplish specific tasks. This type of diagram is particularly useful in modeling the logic behind complex processes, including decisions, loops, and parallel flows.

In this project, Activity Diagrams help visualize how a user interacts with core features of the application. For example:

- The process of recording and analyzing a baby's cry.
- Navigating the app to browse or post items in the marketplace.
- Registering, logging in, and accessing personalized content.

By using Activity Diagrams, we gain a clearer understanding of user behavior, system responses, and potential areas for improvement in the user experience.

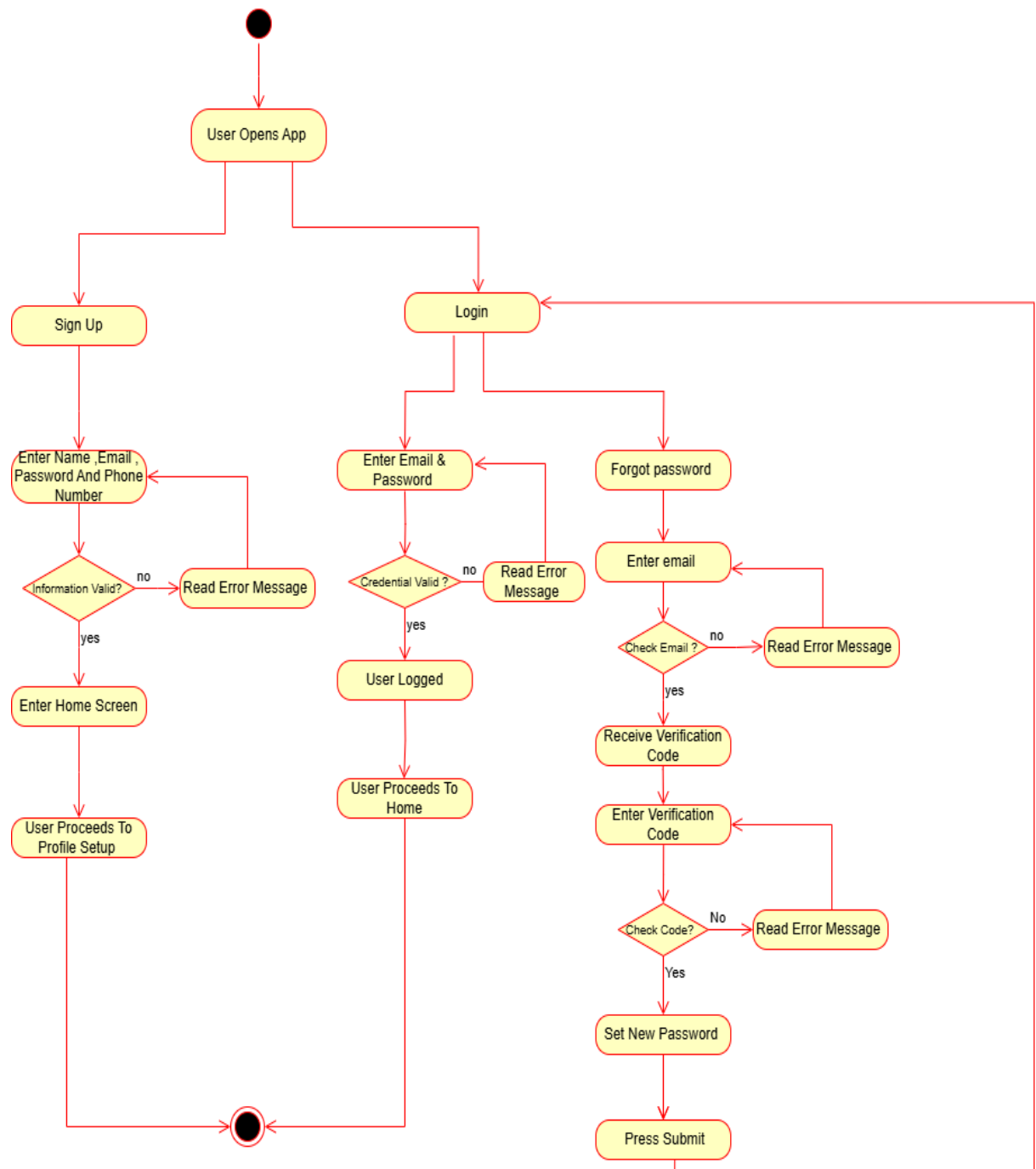


Figure 3.3 activity diagram for Login & Sign Up

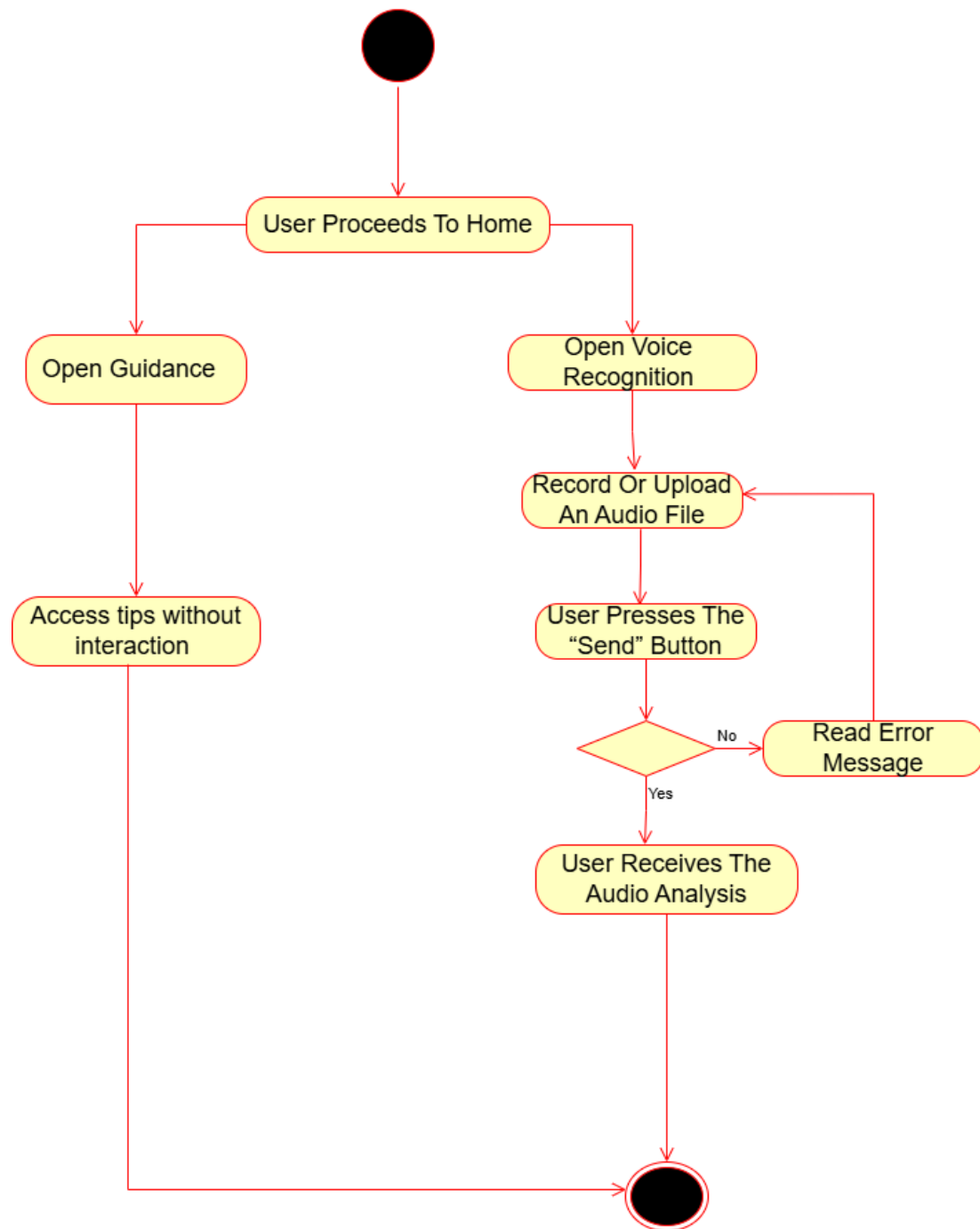


Figure 3.4 activity diagram for Guidance & Voice Recognition

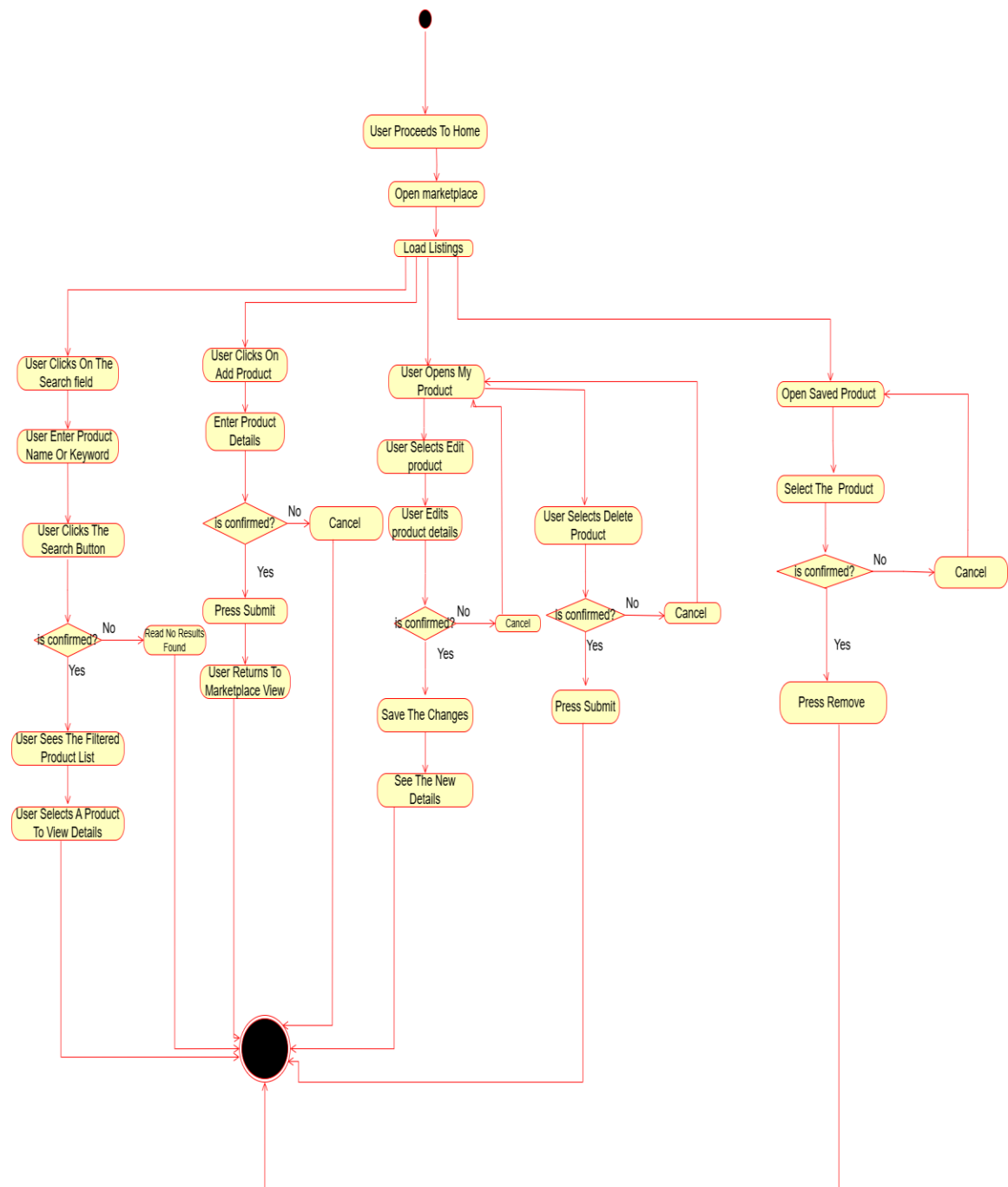


Figure 3.5 activity diagram for Marketplace

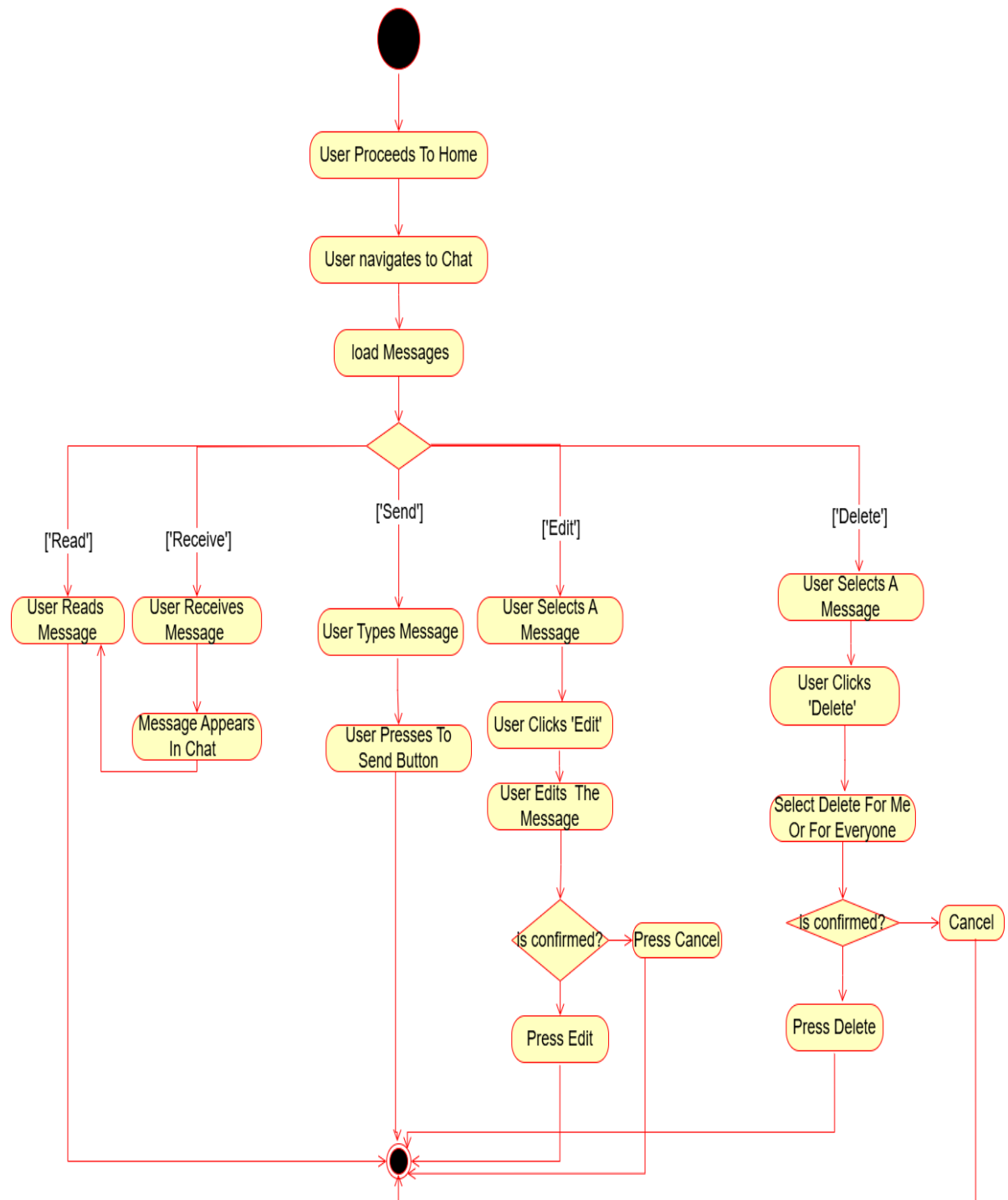


Figure 3.6 activity diagram for Community chat

3.3.4 Motherhood system Sequence diagram:

A **Sequence Diagram** is one of the types of interaction diagrams used in system design. It illustrates **how objects interact with each other in a particular scenario of a use case**, focusing on the sequence of messages exchanged over time.

This diagram emphasizes the **chronological order** of the interactions. Each object or component involved in the process is represented by a **lifeline**, and the messages exchanged are represented by **horizontal arrows** between these lifelines. These messages can include method calls, returns, or signals, and they reflect the flow of information required to accomplish the intended functionality.

Sequence diagrams help in:

- Understanding the flow of logic in a specific use case.
- Clarifying the roles of system components.
- Identifying potential issues or improvements in the interaction process.

In the following diagrams, we present the detailed sequence of actions performed between the user and the system components during specific scenarios in our application.

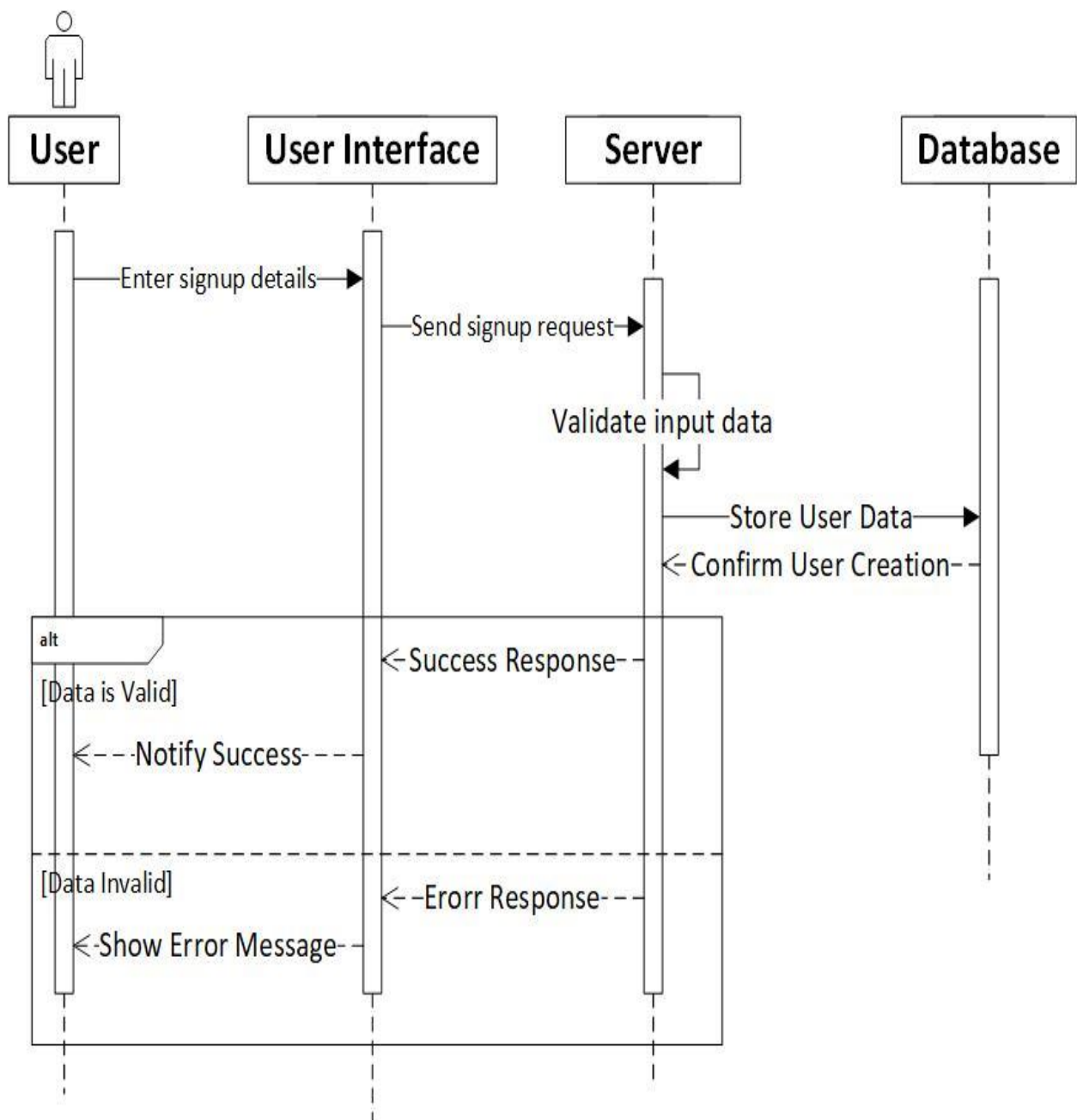


Figure 3.7 Sequence diagram for Sign Up

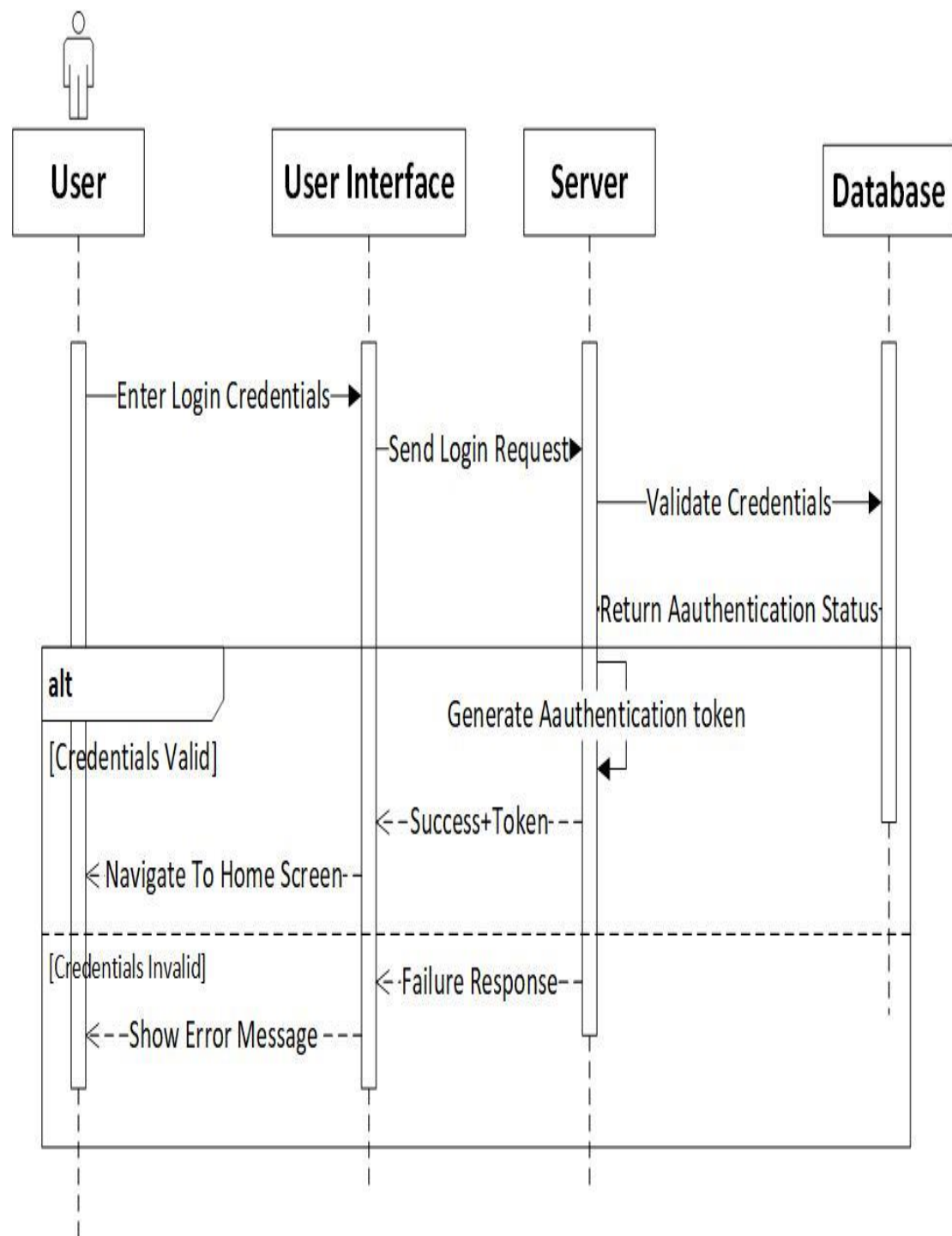


Figure 3.8 diagram for Login

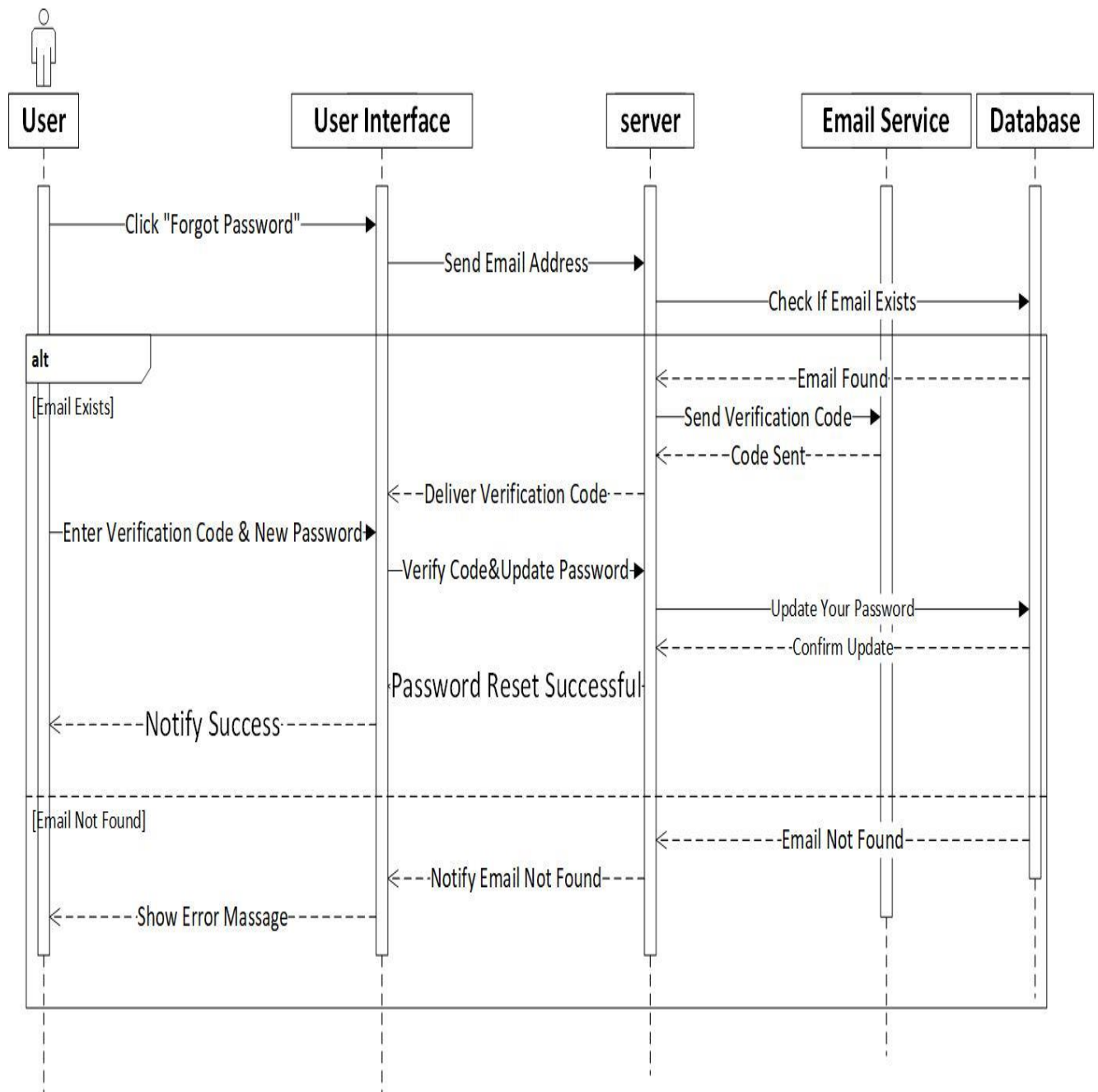


Figure 3.9 Sequence diagram for Forgot Password

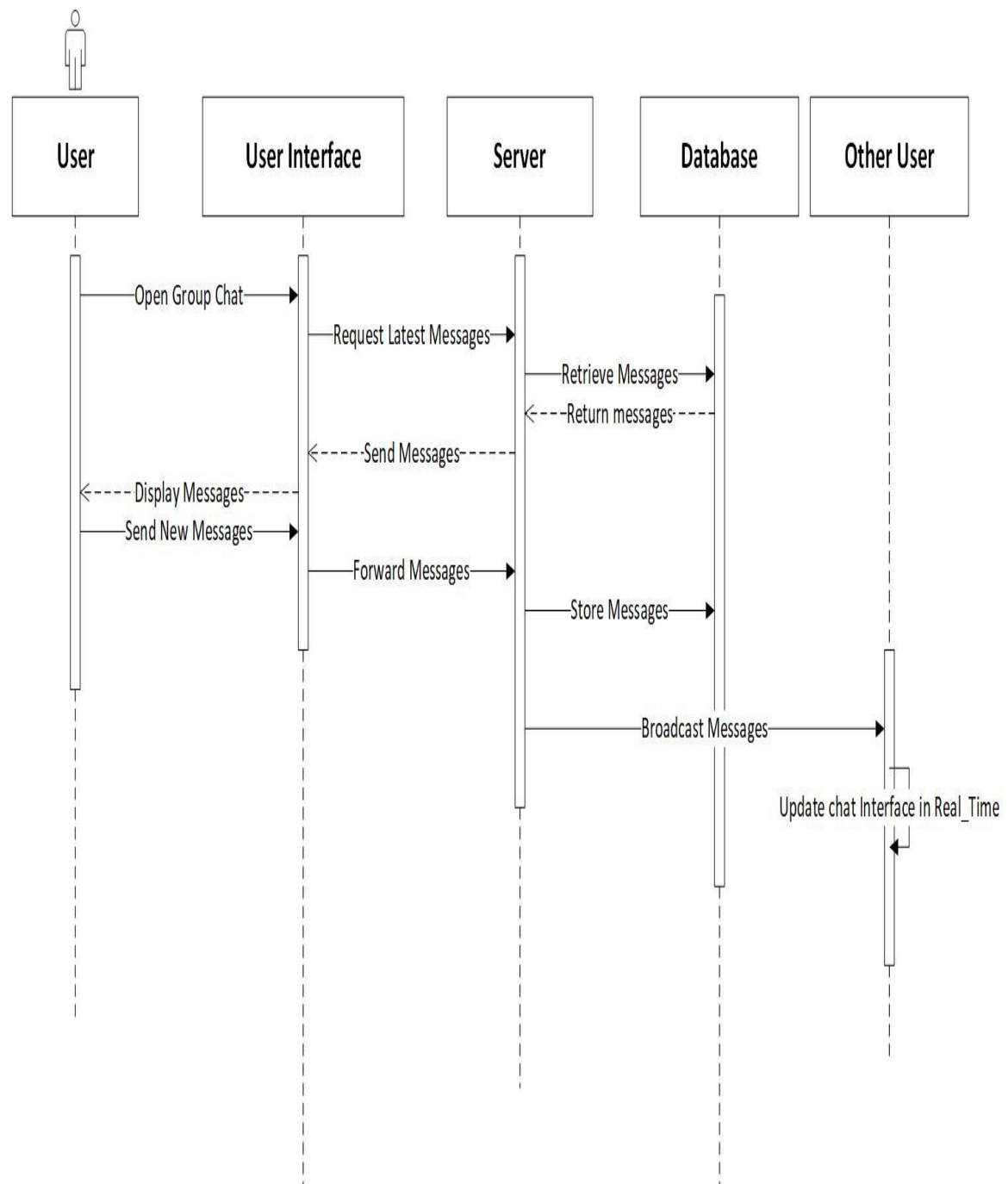


Figure 3.10 Sequence diagram for Group Chat

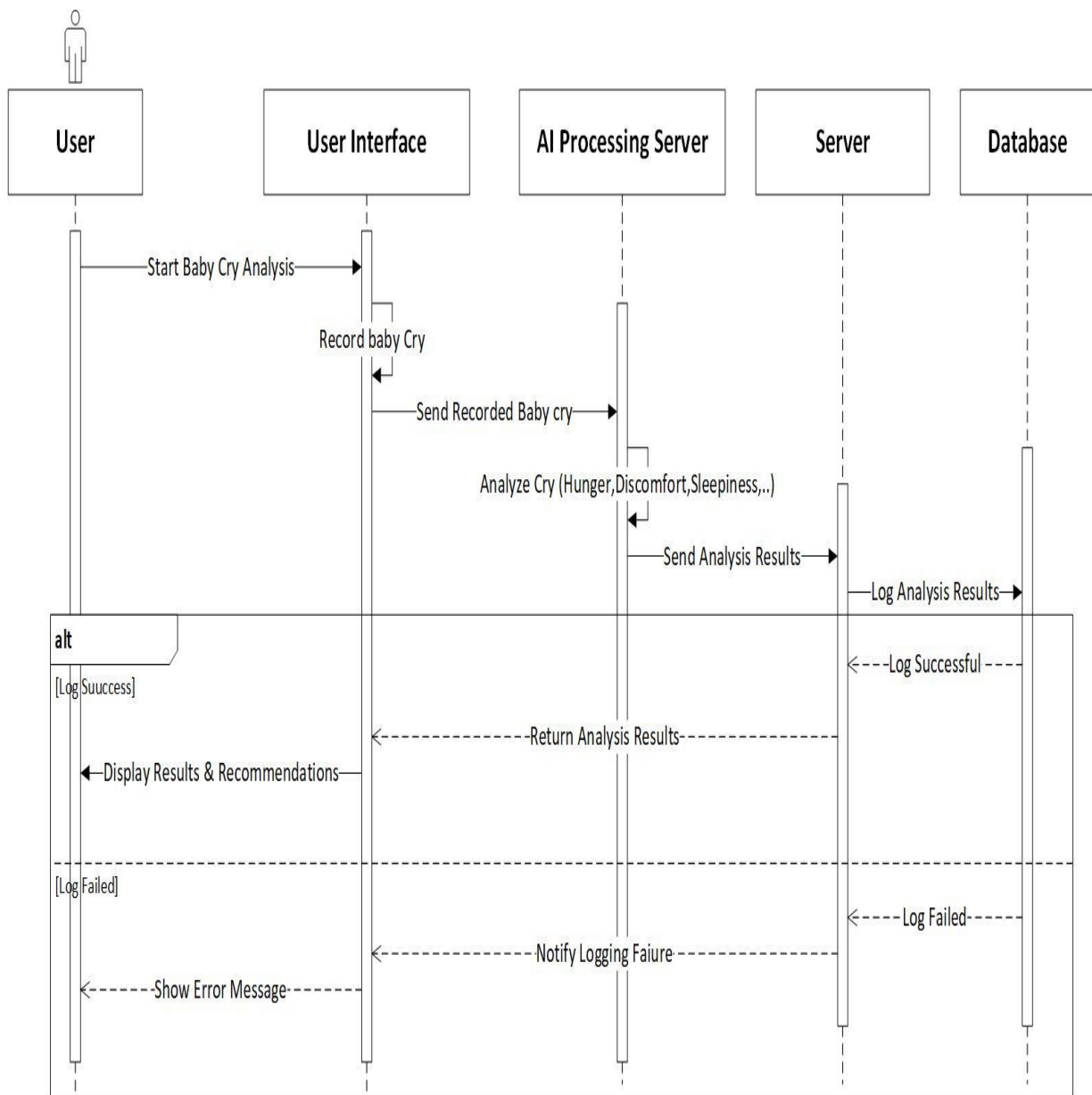


Figure 3.11 Sequence diagram for Baby Cry Analysis

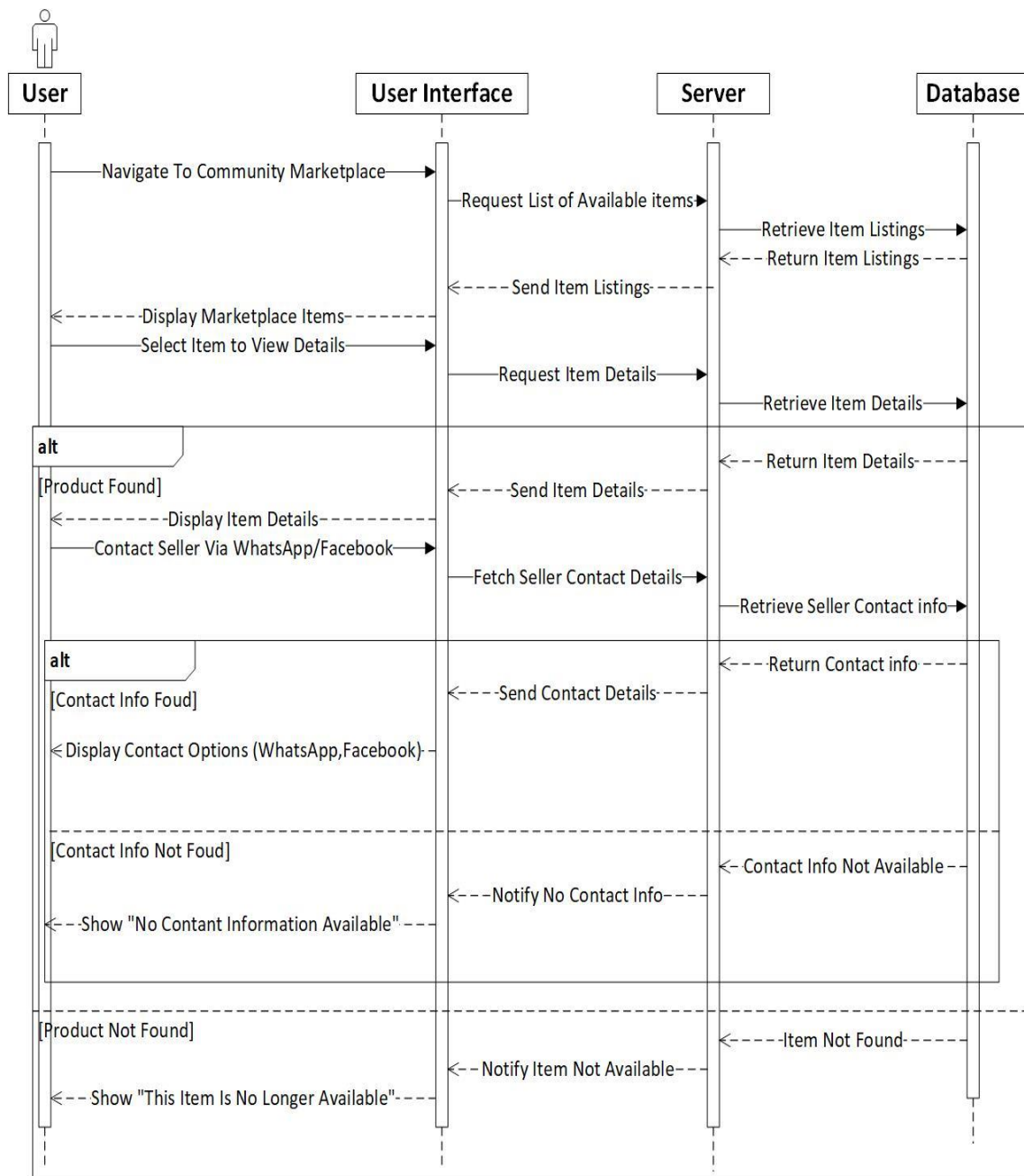


Figure 3.12 Sequence diagram for Marketplace

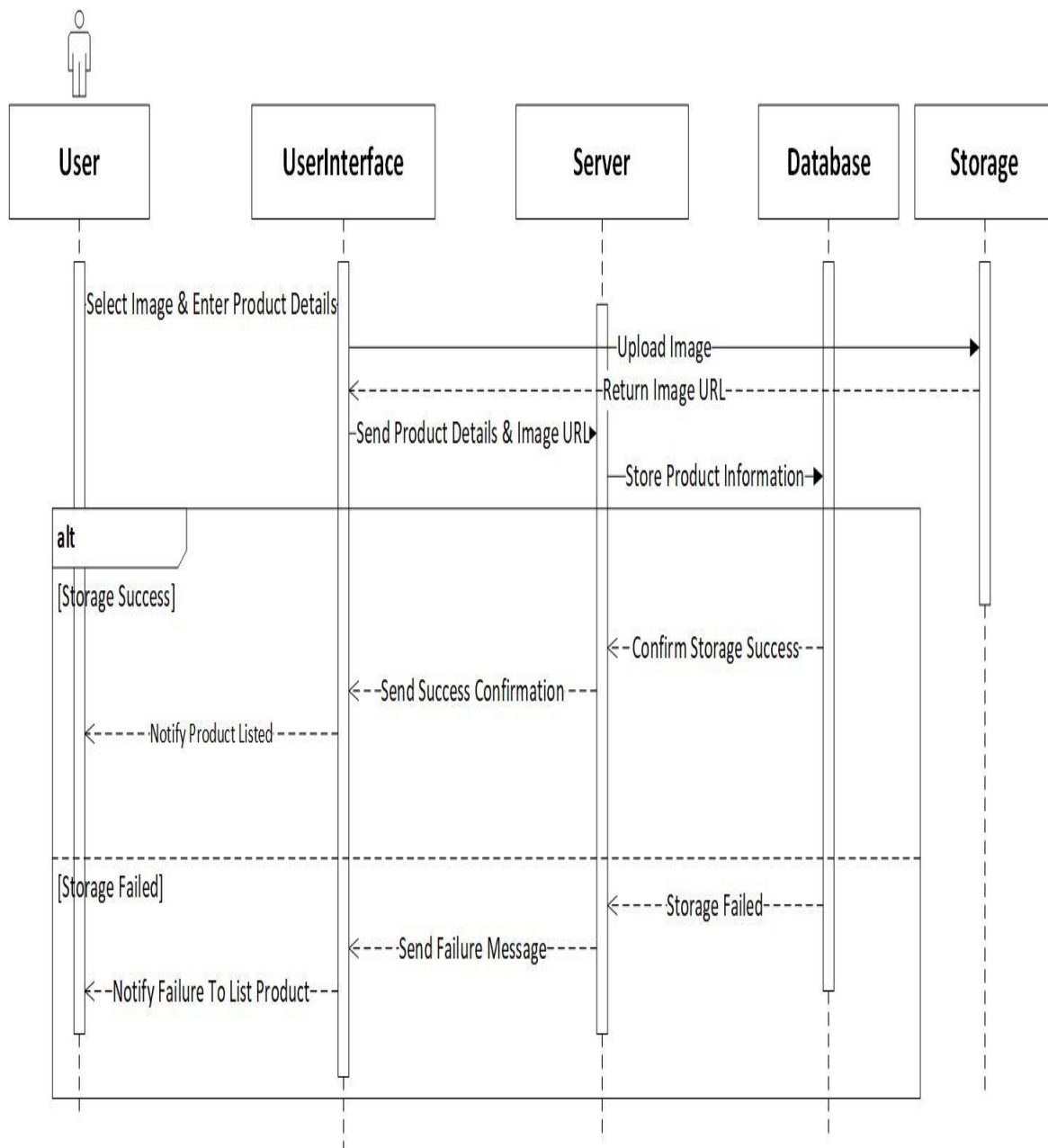


Figure 3.13 Sequence diagram for Add New Product

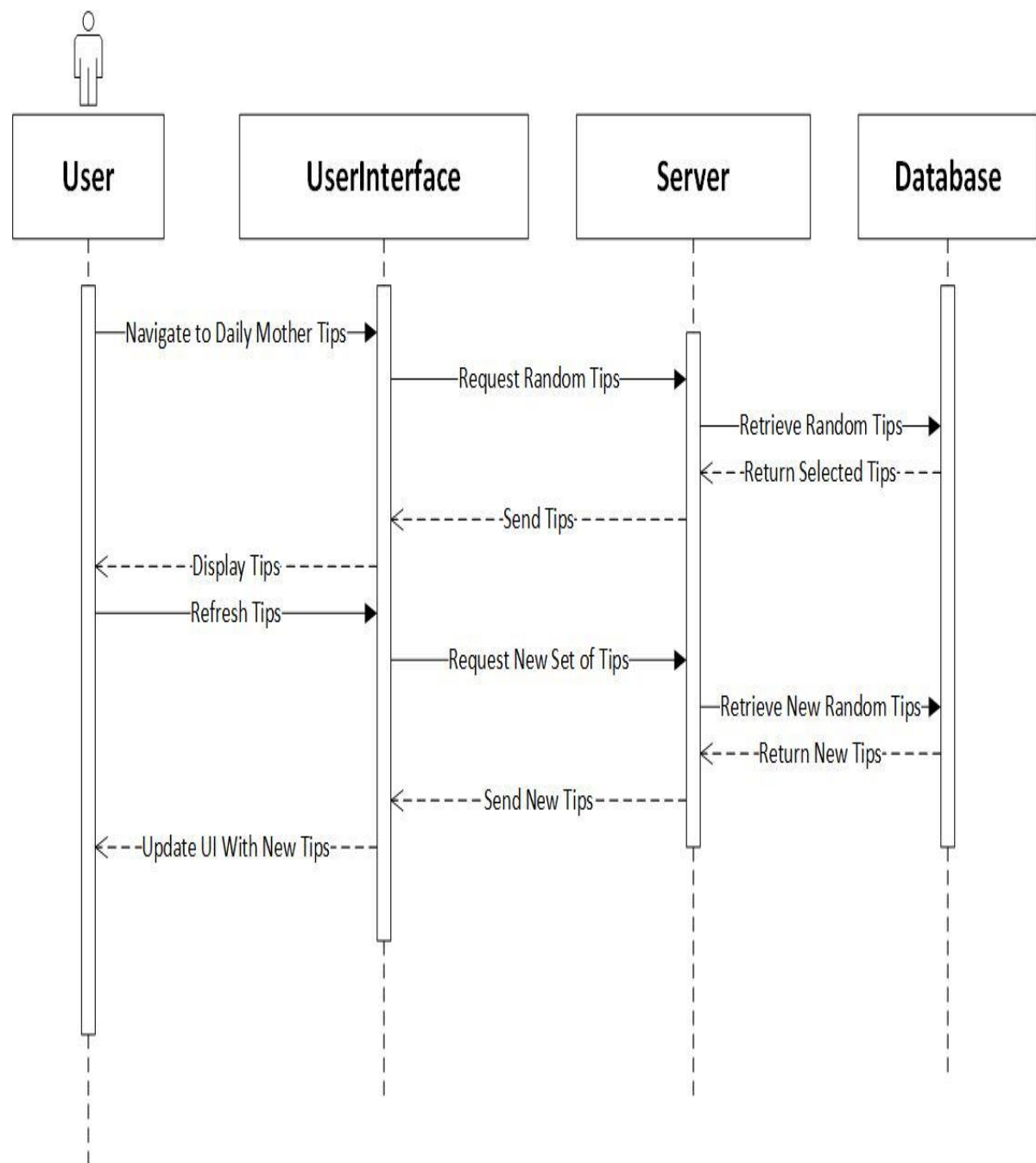


Figure 3.14 Sequence diagram for Daily Tips

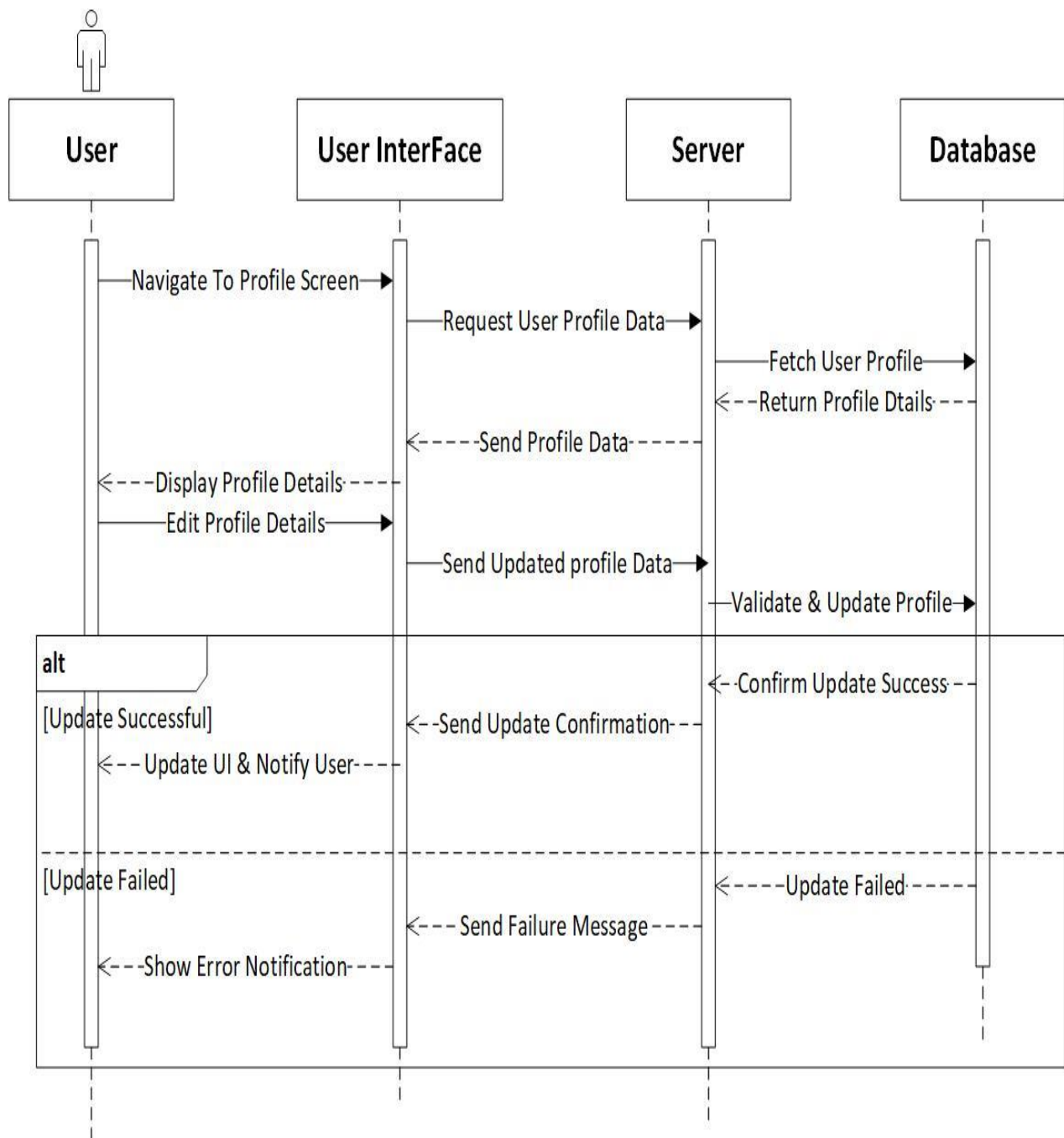


Figure 3.15 Sequence diagram for Profile Screen

3.3.5 ERD Diagram

Entity-Relationship Diagram is a visual representation of the data and their relationships within a system. It shows the entities (like users, products, messages, etc.), their attributes, and how they relate to one another. ERDs are essential tools in database design, helping developers and stakeholders understand how data flows through the system.

3.3.5.1 Entities

1. User: Represents individuals using the system. Users can register, log in, edit their profiles, chat, and interact with marketplace features.

Table 3.2 shows all attributes of user entity

Attribute	Data Type	Description
Id	Nvarchar(450) (primarykey)	A unique identifier for each user to distinguish them in the system.
Username (FirstName,LastName)	Nvarchar(256)	The username used to identify users within the application.
Email	Nvarchar(256)	The email address used for login and communication.
NormalizedEmail	Nvarchar(256)	A standardized version of the email to prevent duplication and improve search efficiency.
PasswordHash	Nvarchar(max)	The encrypted password to secure user data.
PhoneNumber	Nvarchar(max)	The user's phone number for communication or verification

Attribute	Data Type	Description
PhoneNumberConfirmed	Bit	Indicates whether the user has verified their phone number.
TwoFactorEnable	Bit	Specifies if two-factor authentication is enabled for added security.
LockoutEnable	Bit	Determines whether an account can be locked after multiple failed login attempts.
LockoutEnd	Determines whether an account can be locked after multiple failed login attempts.	The time when the lockout period ends if the account is temporarily disabled.
AccessFailedCount	Int	The number of failed login attempts to prevent brute force attack.
ResetCode	Nvarchar(max)	A code for password reset, ensures secure verification before setting a new password.
ResetCodeExpiration	Datetime2	Expiry time for the reset code, prevent unauthorized use after a certain period.
NormalizedUserName	Nvarchar(256)	Ensures case-insensitive user lookup for consistent authentication.
SecurityStamp	Nvarchar(max)	Invalidates old authentication tokens after sensitive account changes.

2. Messages: A message is content exchanged between users in the group chat, allowing them to send and read messages within the conversation.

Table 3.3 shows all attributes of message entity

Attribute	Data Type	Description
Id	Uniqueidentifier (primarykey)	A unique identifier for each message in the system.
Text	Nvarchar(max)	The message content sent by the user.
Timestamp	Datetime2	The time when the message was sent , to maintain chronological order.
IsEdited	Bit	The number of failed login attempts to prevent brute force attack.

3. Message Delete: Deleted messages are messages that the user has removed from their chat. Each user can only see the messages they have deleted.

Table 3.1 shows all attributes of message delete entity

Attribute	Data Type	Description
Id	Unique identifier (Primary key)	A unique identifier for deleted messages to keep track of deletions

4. **Tips:** Tips provide useful information for mothers on baby care, such as advice on sleep, feeding, or common parenting challenges.

Table 3.2 shows all attributes of tips entity

Attribute	Data type	Description
Id	Unique identifier(primary key)	A unique identifier for each tip in the system.
Title	Nvarchar(max)	The title of the tip, providing a brief of its content.
Description	Nvarchar(max)	Detailed information about the tip, offering helpful advice to users.

5. **Baby Records:** This table contains baby cry audio recordings uploaded by users, which are analyzed by an AI model to predict the reason behind the cry.

Table 3.3 shows all attributes of baby records entity

Attribute	Data type	Description
Id	int (primarykey)	A unique identifier for each uploaded baby record.
Filename	Nvarchar(max)	The name of the uploaded file, making it easier to manage and search.
File Path	Nvarchar(max)	The full storage path of the uploaded file on the server.
UploadedAt	Datetime2	The date and time when the file was uploaded, maintaining chronological records.
Prediction	Nvarchar(max)	The AI-generated prediction based on the uploaded file or record.
AIName	Nvarchar(max)	The name of the AI model used for analysis.

6. Products: A product is an item that a user can add and save for sale in the marketplace, including details like name, price, description, and images.

Table 3.4 shows all attributes of products entity

Attribute	Data type	Description
Id	Nvarchar(450) (primarykey)	A unique identifier for each product in the system.
Name	Nvarchar(max)	The product name, making it easier for users to recognize.
Description	Nvarchar(max)	A detailed description of the product, explaining its features and specifications.
State	Nvarchar(max)	The condition of the product (e.g., new, used) to help buyers.
PictureUrl	Nvarchar(max)	The image link of the product for displaying in the interface.
Price	Decimal (18,2)	The price of the product as displayed to users.
PhoneNumber	Nvarchar(max)	The seller's phone number for communication.
WhatsappNumber	Nvarchar(max)	The seller's WhatsApp number as an alternative contact method.
IsLove	Bit	Indicates whether the product has been added to the user's favorites.

3.3.5.2 Relationships

7. User can send multiple Messages, but each Message belongs to exactly one user. A message cannot exist without a user.
8. User can upload multiple baby records; each baby record must be associated with only one user.
9. User can save or add multiple products, and a product can be saved or added by multiple users.
10. user can delete multiple messages from their own view only, but each deleted message must be linked to exactly one user. Deleting a message does not remove it for other users.
11. Each message can have multiple deletion record, and each deleted message must be linked to only one message.
12. User can view multiple tips, and a tip can be linked to multiple users.

However, a tip can exist even if no user views it.

This diagram illustrates the main entities of the system and how they are interconnected.

The core entity is the “Users” table, which holds personal and security-related information. Users can perform various actions, including sending messages, uploading baby cry audio files, adding and saving products, and viewing tips **as shown as in figure 3.16.**

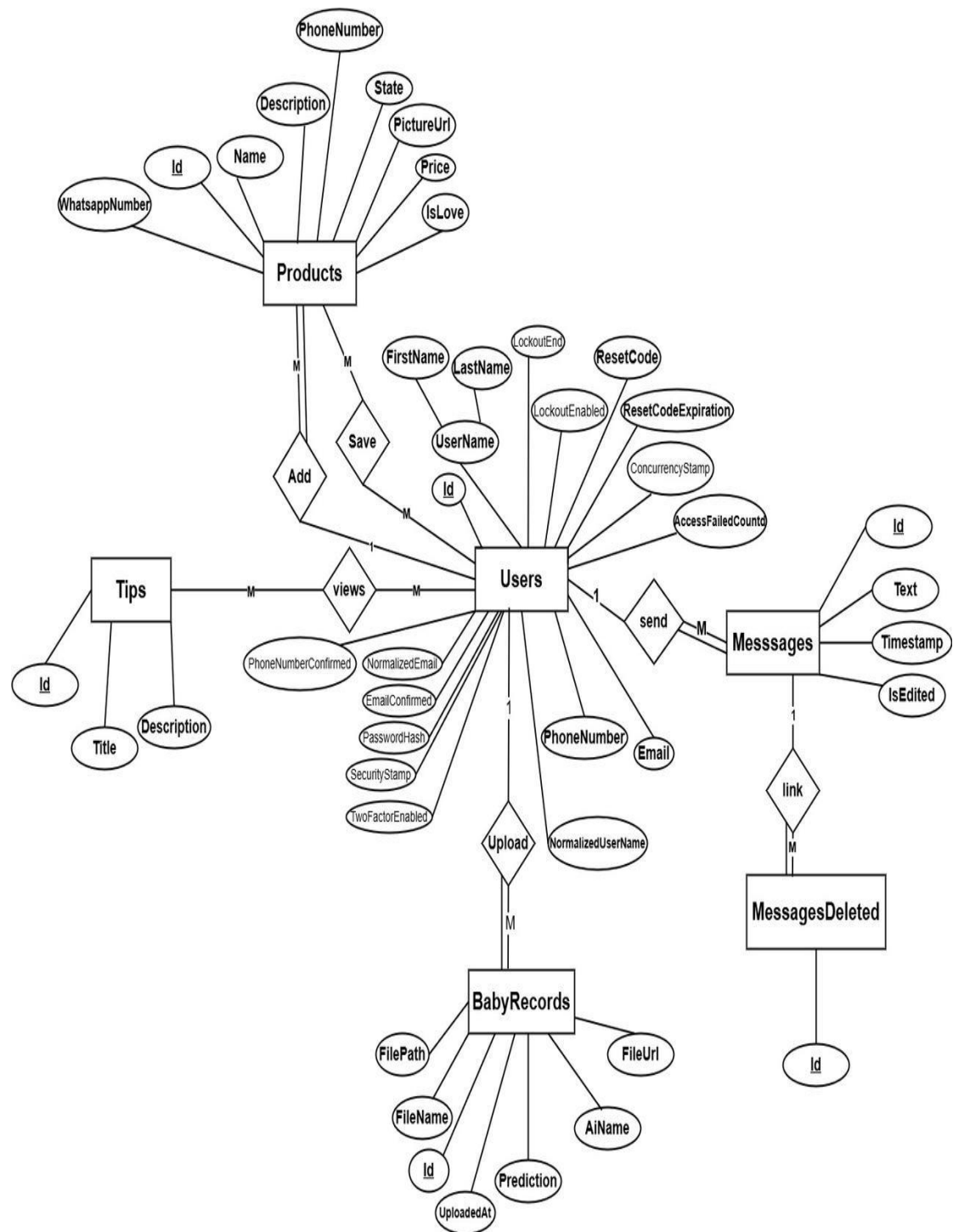


Figure 3.16 ERD diagram

Each user can:

- Send multiple messages, and each message may have an edited status and a timestamp. Messages can also be marked as deleted and linked in the “MessagesDeleted” entity.
- Upload multiple baby cry recordings, where each record includes filename, file path, upload date, and AI-predicted analysis.
- Add or save products, with each product containing details like name, description, price, image, contact numbers, and a flag to indicate whether it’s favorited (IsLove).
- View tips, which consist of a title and description.

The ERD demonstrates the relationships between entities using cardinalities, such as one-to-many and many-to-many, to describe user interactions and system functionalities. It provides a structured overview of the system’s database design.

3.3.6 Schema Diagram

Database schema is the logical structure of a database that defines how data is organized and how the relationships between data are managed. It includes tables, columns, data types, constraints, primary and foreign keys, and the relationships among entities. The schema serves as a blueprint for how information is stored, connected, and retrieved within the system.

The database schema is designed to support a comprehensive system with interconnected entities. The main tables include User, Products, and Baby Records, which manage core data like user profiles, product listings, and baby cry recordings as shown as in figure 3.17.

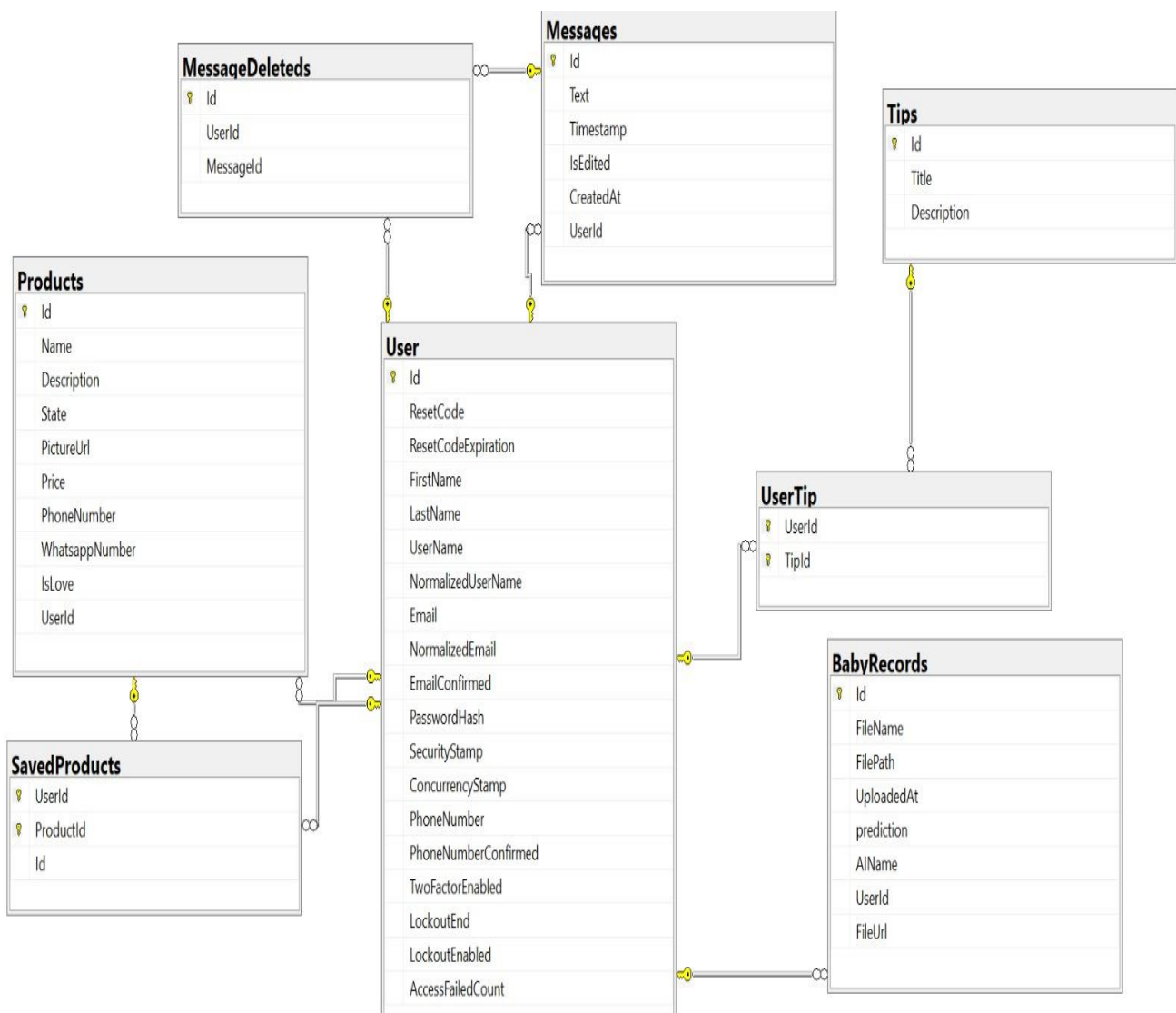


Figure 3.17 Schema Diagram

Many-to-many relationships are handled through linking tables, such as Saved Products, which connects users to their favorite products, and User Tips, which tracks the tips that users have accessed. These tables facilitate personalized user experiences and maintain the integrity of data relationships.

Overall, the schema ensures efficient data organization and management, supporting all key system functionalities like user activities, product interactions, and baby cry analysis.

3.3.7 Class Diagram

A Class Diagram is a type of diagram in the Unified Modeling Language (UML) that represents the static structure of a system. It shows the system's classes, their attributes, methods, and the relationships between the classes.

Key components of a class diagram include:

- **Classes:** Represent objects or entities in the system, with their attributes (data) and operations (functions or methods).
- **Relationships:** Such as associations, generalizations (inheritance), aggregations, and compositions, which define how classes are connected or interact.
- **Inheritance:** Indicates that one class derives from another, inheriting its features.
- **Associations:** Show how instances of one class relate to instances of another.

Class diagrams are primarily used for modeling the design of software systems, helping developers understand and organize code before implementation.

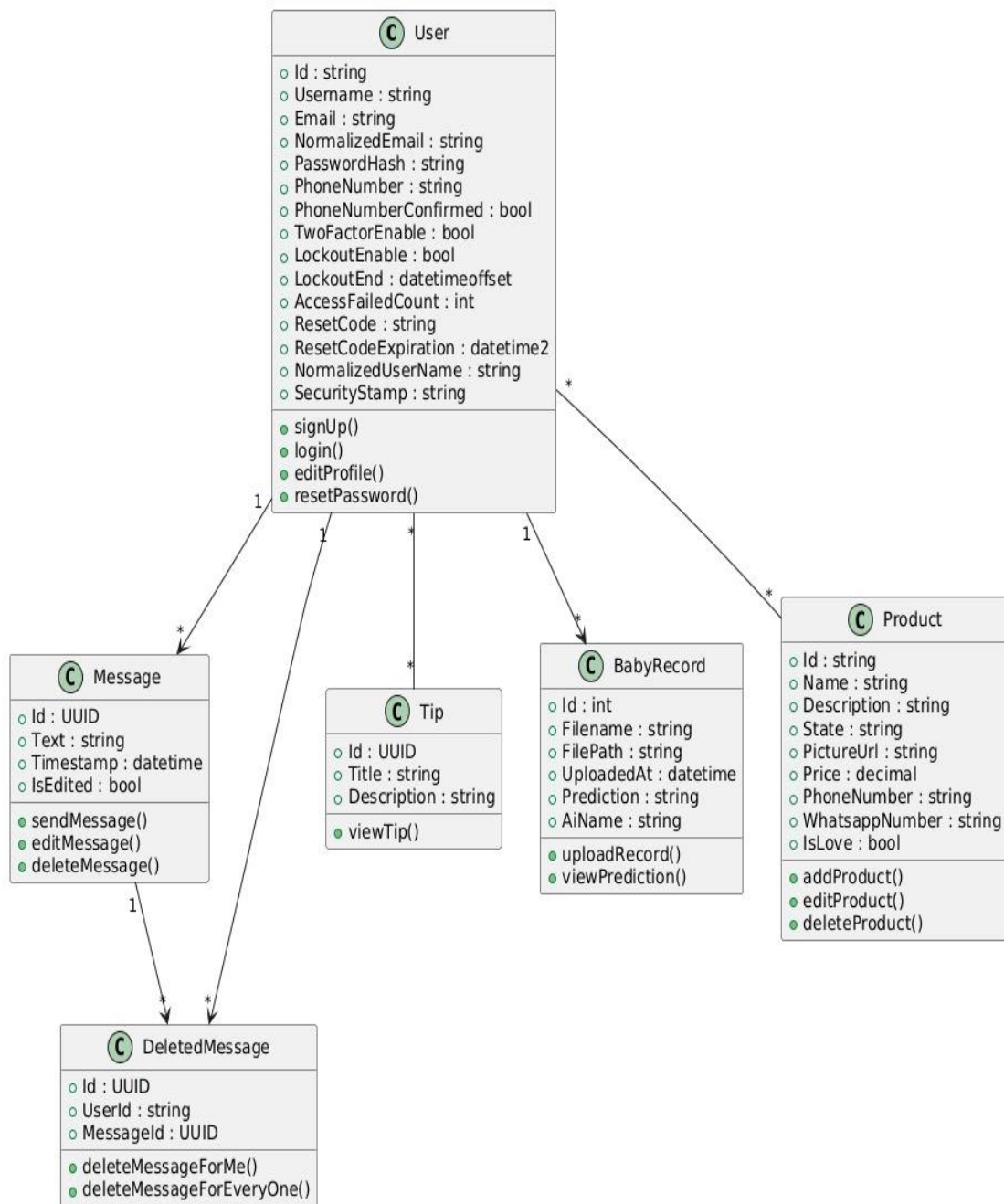


Figure 3.18 Class diagram

Chapter 4: Implementation

Project overview:

This system aims to support new mothers by simplifying childcare through a smart, unified digital platform—unlike traditional, disconnected methods. It consists of multiple integrated modules connected to a single database, ensuring seamless operation across all features.

The system is divided into a mobile application and a website. The mobile app helps mothers manage daily needs on the go, including listing and selling second-hand baby items like toys and clothes directly to other mothers. The web platform offers the same features and also enables community interaction, allowing mothers to connect, share advice, and support each other.

A key feature is the AI-powered voice recognition tool that analyzes recorded baby cries to suggest likely causes such as hunger, burping, tired, belly pain, or discomfort—helping mothers respond more effectively.

Additionally, the platform includes an educational section with expert articles and health tips covering infant care and common developmental-stage illnesses.

By combining AI, peer support, and educational content, this system enhances the motherhood experience and promotes both maternal and infant well-being.

4.1 Web Application

In the following, we will present and identify the modules of the Motherhood system as implemented in the web application.

4.1.1 Signup Screen

On this page, the user can be authenticated regularly performed by entering an email, username and password combination **as illustrated in figure 4.1.**

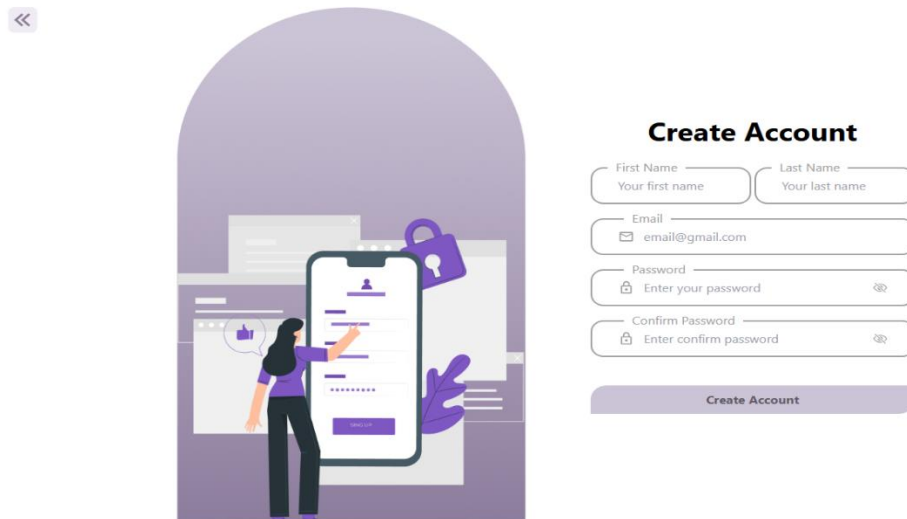


Figure 4.1 Signup Screen

This function is used to verify the email, username and password from the API while signing in to the system as illustrated in Figure 4.2

```

var existingUser = await _userManager.FindByEmailAsync(registerDto.Email);
if (existingUser != null)
{
    return BadRequest(new
    {
        email = "This email is already taken",
        password = (string?)null
    });
}
var passwordValidator = new PasswordValidator<AppUser>();
var passwordResult = await passwordValidator.ValidateAsync(_userManager, null,
registerDto.Password);
if (!passwordResult.Succeeded)
{
    return BadRequest(new
    {
        email = (string?)null,
        password = "Password must be at least 8 characters, include uppercase,
number"
    });
}
var user = new AppUser()
{
    Email = registerDto.Email,
    UserName = registerDto.Email.Split('@')[0],
    PhoneNumber = registerDto.PhoneNumber,
    FirstName = registerDto.FirstName,
    LastName = registerDto.LastName,
};
var result = await _userManager.CreateAsync(user, registerDto.Password);
if (!result.Succeeded)
{
    return BadRequest(new
    {
        email = "Error creating user",
        password = (string?)null
    });
}

```

Figure 4.2 Signup Function

4.1.2 Login Screen

On this page, the user is authenticated by entering a valid email and password combination, as illustrated in Figure 4.3.

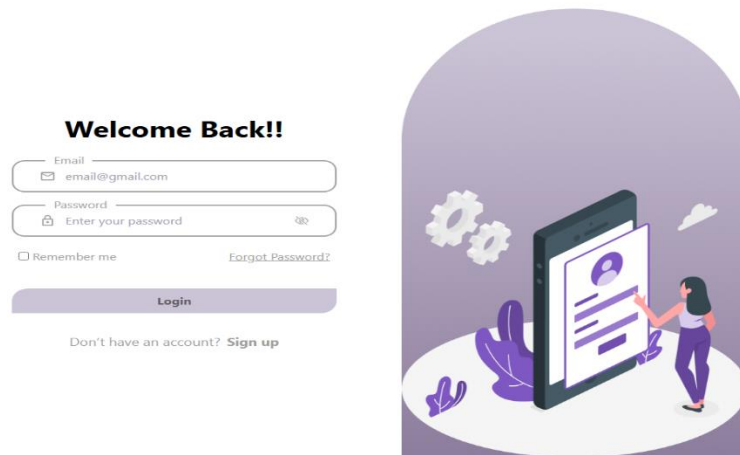


Figure 4.3 Login Screen

This function is responsible for verifying the email and password via the API during the login process, as illustrated in Figure 4.4.

```
var user = await _userManager.FindByEmailAsync(loginDto.Email);
if (user == null)
{
    return Unauthorized(new
    {
        email = "Email not found",
        password = (string?)null
    });
}
var result = await _signInManager.CheckPasswordSignInAsync(user, loginDto.Password,
false);
if (!result.Succeeded)
{
    return Unauthorized(new
    {
        email = (string?)null,
        password = "Incorrect password"
    });
}
var token = await _tokenServices.CreateToken(user, _userManager);
return Ok(new UserDto()
{
    Email = user.Email,
    Token = token,
    Id = user.Id,
    FirstName = user.FirstName,
    LastName = user.LastName,
});
```

Figure 4.4 Login Function

4.1.3 Tips Screen

This screen displays the tips presented to the user, as illustrated in Figure 4.5.

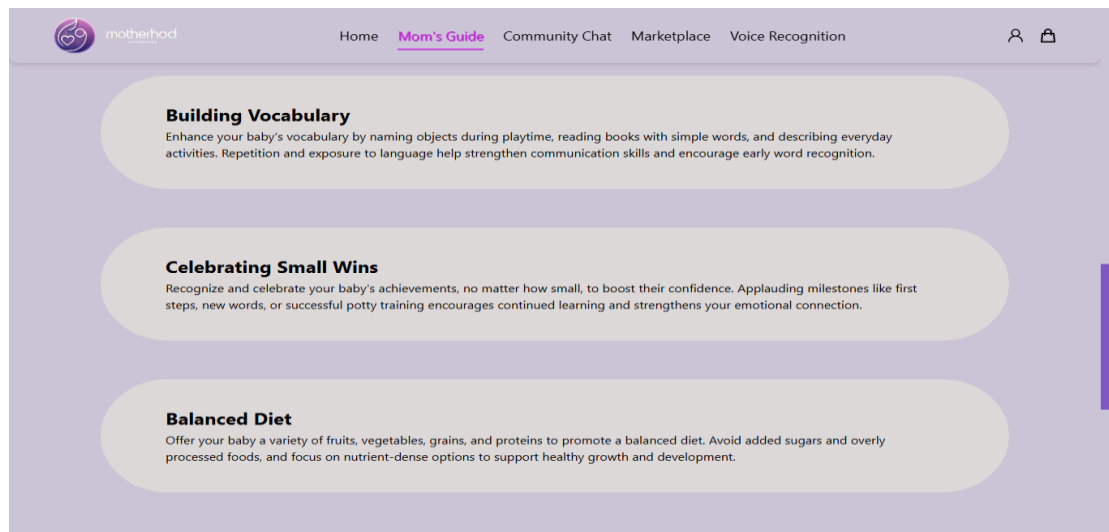


Figure 4.5 Tips Screen

This screen displays the data retrieved from the API and presented to the user, as illustrated in Figure 4.6.

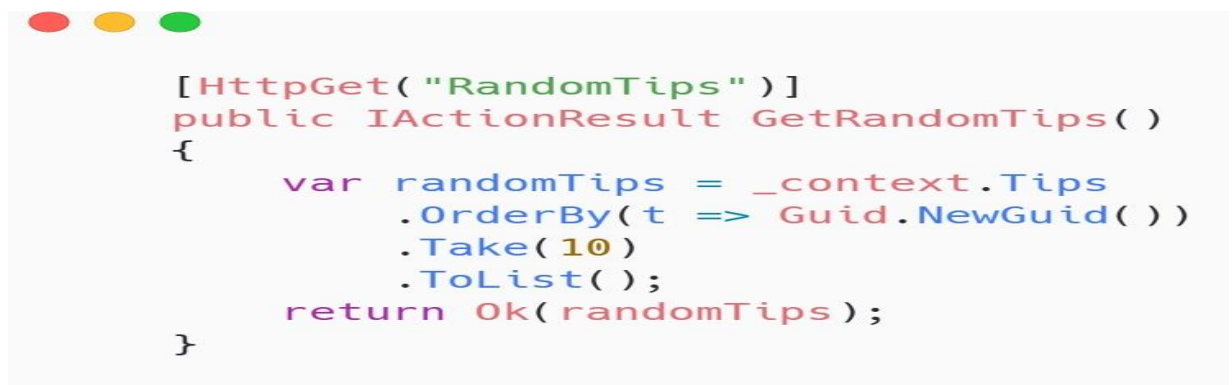


Figure 4.6 Tips Function

4.1.4 Chat Screen

This screen displays the chat interface, allowing users to send and receive messages in real time, as illustrated in Figure 4.7.

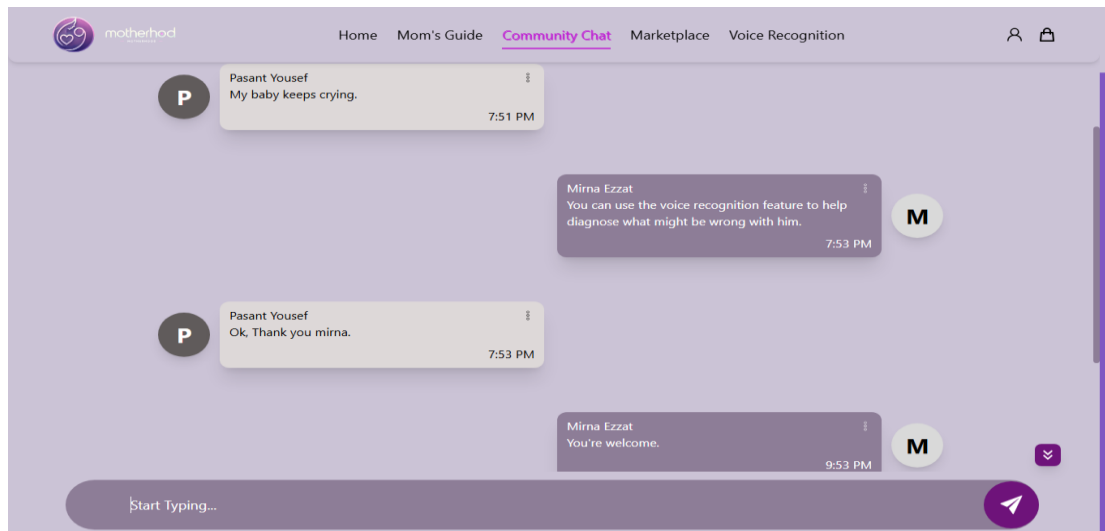


Figure 4.7 Chat Screen

This screen displays the messages exchanged through the chat API, enabling real-time communication between users, **as illustrated in**

Figure 4.8.

```
[HttpPost]
[Authorize(AuthenticationSchemes = "Bearer")]
public async Task<IActionResult> Create([FromBody] MessageDto messageDto)
{
    var userId = User.FindFirstValue(ClaimTypes.NameIdentifier);
    var message = new Message
    {
        Text = messageDto.Text,
        Timestamp = DateTime.UtcNow,
        UserId = userId
    };
    _context.Messages.Add(message);
    await _context.SaveChangesAsync();
    return CreatedAtAction(nameof(GetById), new { id = message.Id },
message);
}
```

Figure 4.8 Chat Function

This code segment illustrates how the system deletes chat messages by identifying the selected message, verifying permissions, and removing it from the conversation history, **as illustrated in Figure 4.9.**



```
[HttpDelete("for-me/{id}")]
public async Task<IActionResult> DeleteForMe(Guid id)
{
    var userId = User.FindFirstValue(ClaimTypes.NameIdentifier);
    if (await _context.MessageDeletedds.AnyAsync(md => md.MessageId == id && md.UserId == userId))
        return BadRequest("Already deleted.");

    _context.MessageDeletedds.Add(new MessageDeleted { UserId = userId, MessageId = id });
    await _context.SaveChangesAsync();
    return Ok(new { message = "Deleted for you only." });
}

[HttpDelete("{id}")]
public async Task<IActionResult> Delete(Guid id)
{
    var userId = User.FindFirstValue(ClaimTypes.NameIdentifier);
    var message = await _context.Messages.FirstOrDefaultAsync(m => m.Id == id && m.UserId ==
userId);
    if (message == null) return NotFound();

    _context.MessageDeletedds.RemoveRange(_context.MessageDeletedds.Where(md => md.MessageId ==
id));
    _context.Messages.Remove(message);
    await _context.SaveChangesAsync();
    return Ok(new { message = "Deleted for everyone." });
}
```

Figure 4.9 Deleting Message Function

4.1.5 Voice Recognition Screen

This screen displays the voice recognition feature, allowing users to record and analyze audio input using AI, **as illustrated in Figure 4.10.**

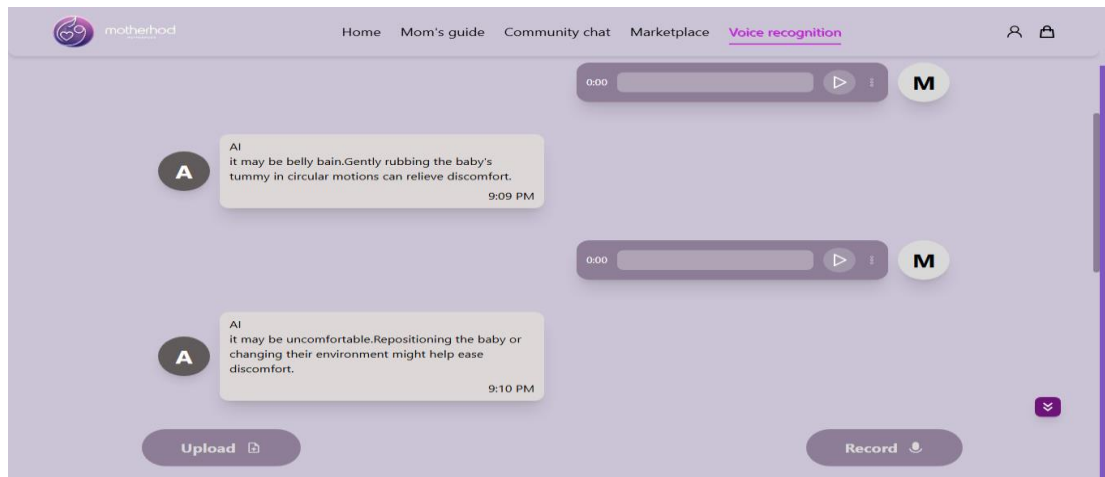


Figure 4.10 Voice Recognition Screen

This code segment demonstrates the logic implemented to process the audio input using Python, which sends the data to the voice recognition API and handles the interpreted results, as illustrated in Figure 4.11.

```

@app.route('/predict', methods=['POST'])
def predict_route():
    try:
        file = request.files.get('file')
        if not file:
            return jsonify({"error": "No audio file provided."}), 400

        file_ext = os.path.splitext(file.filename)[1].lower()
        temp_path = os.path.join("uploads", file.filename)
        file.save(temp_path)

        supported_formats = ['.wav', '.webm', '.mp4', '.mp3', '.m4a', '.ogg', '.aac', '.flac']
        if file_ext not in supported_formats:
            return jsonify({"error": f"Unsupported file format: {file_ext}"}), 400
        if file_ext != ".wav":
            try:
                sound = AudioSegment.from_file(temp_path)
                wav_path = temp_path.replace(file_ext, ".wav")
                sound.export(wav_path, format="wav")
                os.remove(temp_path)
                temp_path = wav_path
            except Exception as e:
                return jsonify({"error": f"Conversion error: {str(e)}"}), 500
        predict, _ = predict_audio(temp_path)

        return jsonify({
            "prediction": predict,
        })

```

Figure 4.11 Voice Recognition Python Code

This screen displays the results processed by the voice recognition API, which analyzes the recorded audio and returns the interpreted data, as illustrated in Figure 4.12.

```
var client = _httpClientFactory.CreateClient();
using var form = new MultipartFormDataContent();
using var fileStream = new FileStream(filePath, FileMode.Open, FileAccess.Read);
using var streamContent = new StreamContent(fileStream);
streamContent.Headers.ContentType = new MediaTypeHeaderValue("application/octet-stream");
form.Add(streamContent, "file", uniqueFileName);

var flaskResponse = await client.PostAsync("http://127.0.0.1:5000/predict", form);
var predictionJson = await flaskResponse.Content.ReadAsStringAsync();

if (flaskResponse.IsSuccessStatusCode)
{
    var predictionObj = JsonSerializer.Deserialize<JsonElement>(predictionJson);

    if (predictionObj.TryGetProperty("prediction", out JsonElement predictionElement))
    {
        predictedClass = predictionElement.GetString() ?? "No Prediction";
    }
}
```

Figure 4.12 Voice Recognition Function

4.1.6 Marketplace Screen

This screen displays the marketplace interface, where users can browse, list, and purchase second-hand items, as illustrated in Figure 4.13.

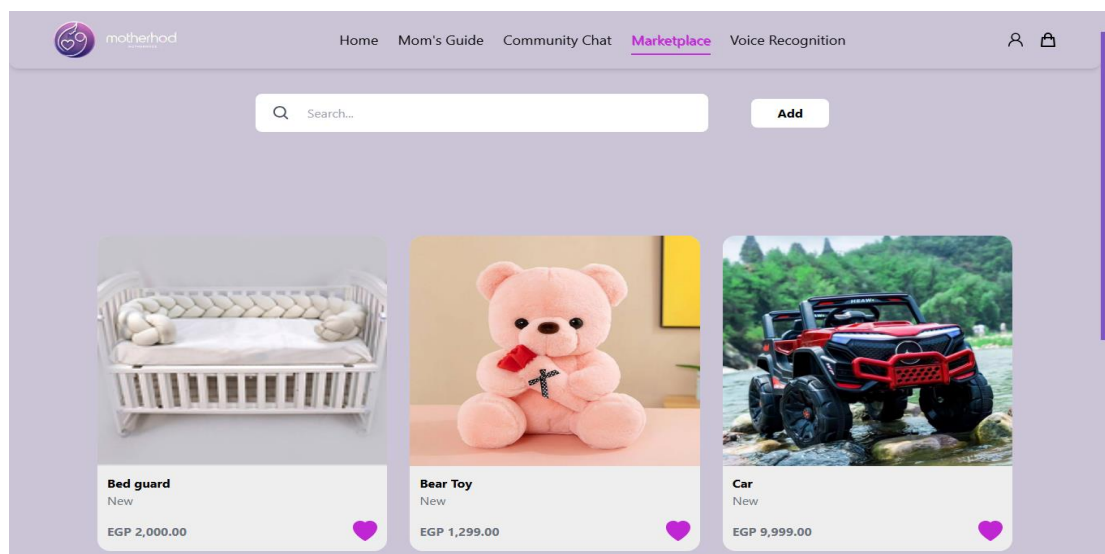


Figure 4.13 Marketplace Screen

This screen displays the data retrieved from the marketplace API, including item listings and transaction details, as illustrated in Figure 4.14.

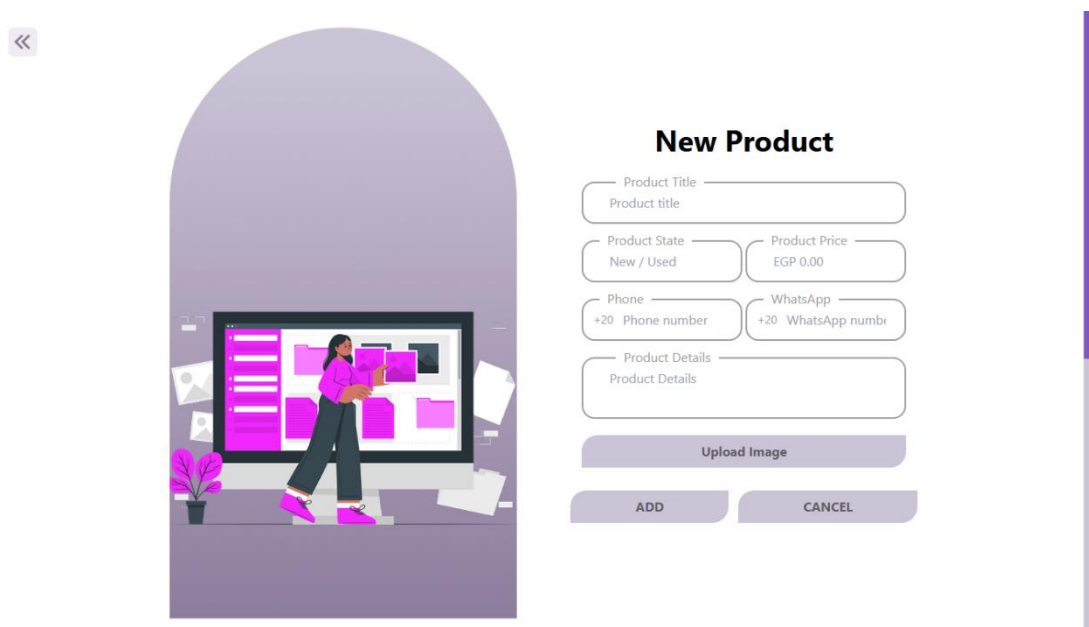
```
[HttpGet]
public async Task<ActionResult<Pagination<ProductToReturnDto>>> GetProducts([FromQuery] AllProductsPram
pram)
{
    var products = await _productService.GetProductsAsync(pram);
    var count = await _productService.GetCountAsync(pram);
    var data = products.Select(product => new ProductToReturnDto
    {
        Id = product.Id,
        Name = product.Name,
        Description = product.Description,
        PictureUrl = product.PictureUrl,
        State = product.State,
        Price = product.Price,
        UserName = product.User?.UserName ?? string.Empty,
        PhoneNumber = product.PhoneNumber,
        WhatsappNumber = product.WhatsappNumber
    }).ToList();

    return Ok(new Pagination<ProductToReturnDto>(pram.PageIndex, pram.PageSize, count, data));
}
```

Figure 4.14 Marketplace Function

4.1.7 Add Product Screen

This screen allows users to add new products to the marketplace by entering item details and uploading image, as illustrated in Figure 4.15.



The image is a composite. On the left, there is an illustration of a person with dark hair, wearing a pink long-sleeved shirt and dark trousers, standing in front of a large computer monitor. The monitor displays a web application with a grid of pink cards. The person is holding a pink folder. To the right of the illustration is a vertical purple bar. On the right side of the image is a 'New Product' form. The form has a title 'New Product' in bold. It contains several input fields: 'Product Title' (with a placeholder 'Product title'), 'Product State' (with a placeholder 'New / Used'), 'Product Price' (with a placeholder 'EGP 0.00'), 'Phone' (with a placeholder '+20 Phone number'), 'WhatsApp' (with a placeholder '+20 WhatsApp number'), and 'Product Details' (with a placeholder 'Product Details'). Below these fields are three buttons: 'Upload Image', 'ADD', and 'CANCEL'.

Figure 4.15 Add Product Screen

This screen interacts with the marketplace API to submit new product data, including item details and images, for listing, as **illustrated in Figure 4.16**.

```
var product = new Product
{
    Name = productDto.Title,
    Description = productDto.Details,
    PictureUrl = pictureUrl,
    State = productDto.State,
    Price = productDto.Price,
    IsLove = productDto.IsLove,
    PhoneNumber = productDto.PhoneNumber,
    WhatsappNumber = productDto.WhatsappNumber,
    UserId = userId
};
var createdProduct = await
_productApi.CreateProductAsync(product);
{
    return BadRequest(new ApiResponse(400, "Problem creating product"
));
}
```

Figure 4.16 Add Product Function

4.1.8 Forget Password

This screen allows users to reset their password by entering their registered email address, as **illustrated in Figure 4.17**.

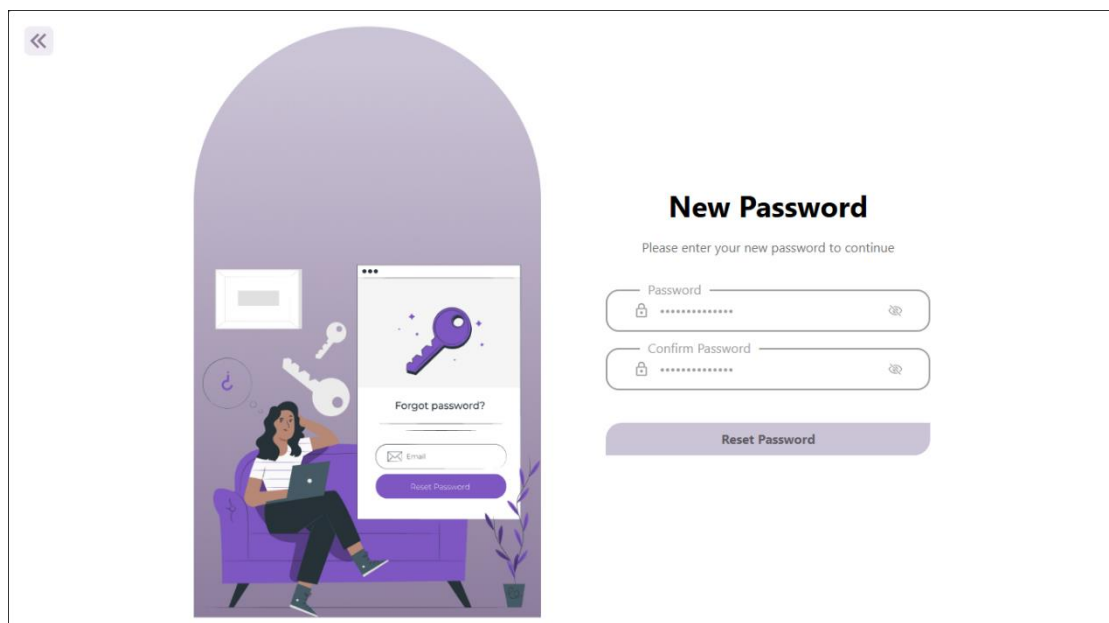


Figure 4.17 Forget Password Screen

This screen interacts with the authentication API to initiate the password reset process by submitting the user's registered email address and allowing them to set a new password, **as illustrated in Figure 4.18.**

```
var user = await _userManager.FindByEmailAsync(forgotPasswordDto.Email);
if (user == null) return NotFound(new { message = "User not found" });
var resetCode = new Random().Next(100000, 999999).ToString();
user.ResetCode = resetCode;
user.ResetCodeExpiration = DateTime.UtcNow.AddMinutes(5);
var updateResult = await _userManager.UpdateAsync(user);
if (!updateResult.Succeeded) return BadRequest(new { message = "Error updating user's reset
code" });
var subject = "Password Reset Code";
var body = $"<p>Your password reset code is: <strong>{resetCode}</strong></p><p>This code is
valid for 5 minutes.</p>";
await emailService.SendEmailAsync(user.Email, subject, body);
return Ok(new { message = "Reset code sent to your email successfully." });
}
```

Figure 4.18 Forget Password Function

4.2 Mobile Application

In the following, we will present and identify the modules of the Motherhood system as implemented in the mobile application.

4.2.1 Signup Screen

This screen allows users to create a new account by entering their personal information and credentials, **as illustrated in Figure 4.19.**

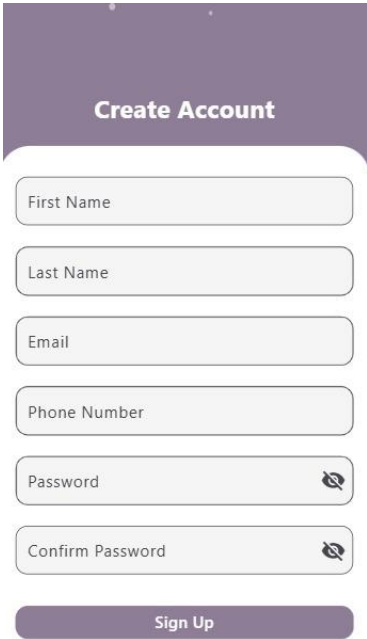


Figure 4.19 Signup Screen

4.2.2 Login Screen

This screen enables users to access their accounts by entering their email and password, **as illustrated in Figure 4.20.**

The login screen features a purple header with the 'mother Hood' logo. Below the header, the word 'Login' is centered. There are two input fields: 'Email' and 'Password'. The 'Password' field has a toggle icon on the right. Below the 'Password' field, there is a checkbox labeled 'Remember me' and a link labeled 'Forgot Password?'. A purple 'Login' button is positioned below these elements. At the bottom, there is a link that says 'Don't have an Account? Register'.

Figure 4.20 Login Screen

4.2.3 Forget Password Screen

This screen allows users to initiate the password recovery process by submitting their registered email address, **as illustrated in Figure 4.21.**

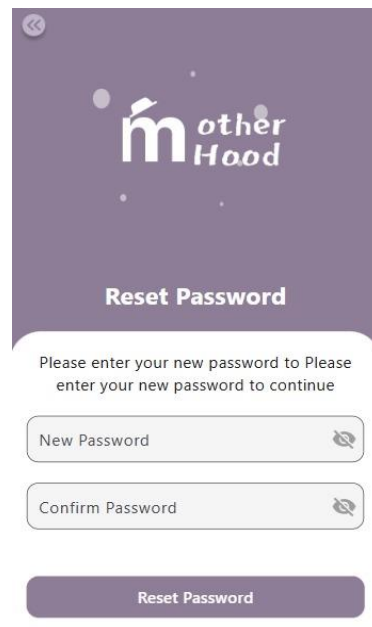
The reset password screen has a purple header with the 'mother Hood' logo. Below the header, the text 'Reset Password' is centered. A message reads: 'Please enter your new password to Please enter your new password to continue'. There are two input fields: 'New Password' and 'Confirm Password'. Both fields have toggle icons on the right. A purple 'Reset Password' button is located at the bottom of the form.

Figure 4.21 Forget Password Screen

4.2.4 Tips Screen

This screen provides users with parenting and health-related tips tailored to their needs, **as illustrated in Figure 4.22.**

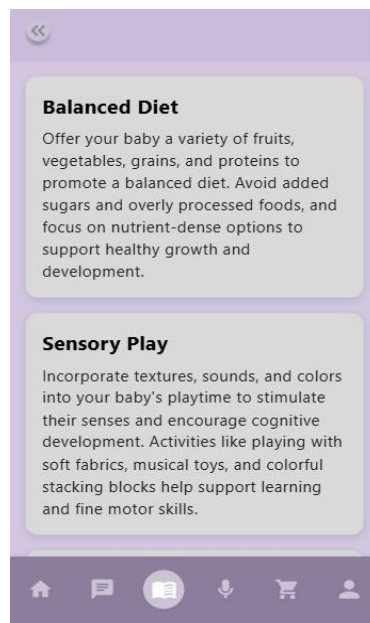


Figure 4.22 Tips Screen

4.2.5 Chat Screen

This screen enables users to engage in real-time conversations within the community, allowing them to share experiences and support, **as illustrated in Figure 4.23.**

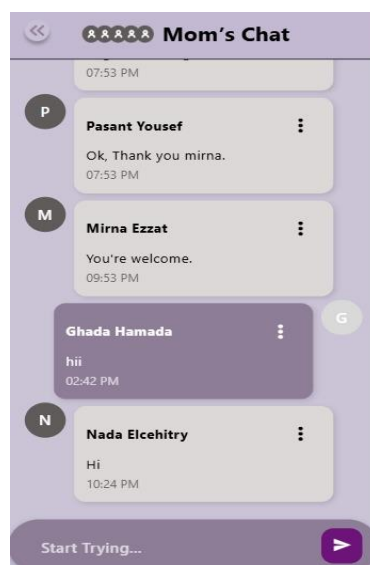


Figure 4.23 Chat Screen

4.2.6 Marketplace Screen

This screen allows users to browse, search, and view products listed by other community members, **as illustrated in Figure 4.24.**

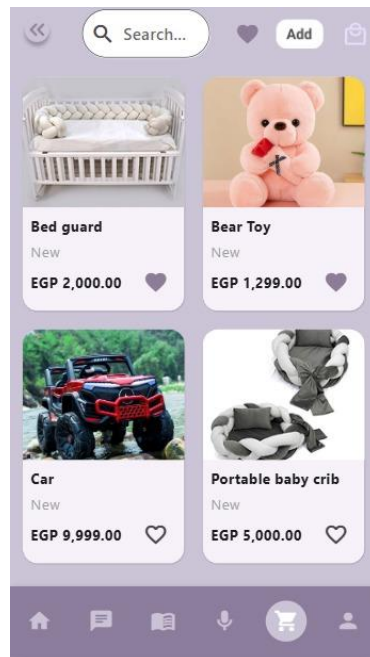


Figure 4.24 Marketplace Screen

4.2.7 Add Product Screen

This screen enables users to list new products by entering item details and uploading images, interacting with the marketplace API to submit the data, **as illustrated in Figure 4.25.**

A screenshot of a mobile application's 'Add Product' screen. The screen has a purple header with a decorative graphic. Below the header, there are several input fields: 'Product Title', 'Product State', 'Product Price', 'Phone', 'WhatsApp', and a larger 'Details' field. Below these fields, there is a purple button labeled 'Upload Image'. At the bottom, there are two buttons: 'Save' and 'Cancel'.

Figure 4.25 Add Product Screen

4.2.8 Voice Recognition Screen

This screen presents the voice recognition feature, enabling users to record and analyze audio input through AI technologies, **as illustrated in Figure 4.26.**

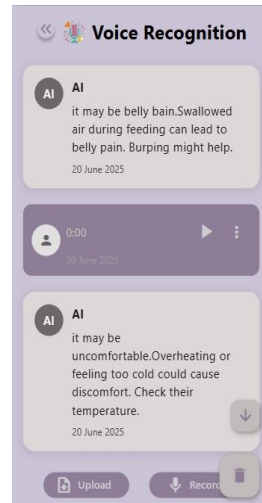


Figure 4.26 Voice Recognition Screen

4.3 Conclusion

The implementation of the Motherhood project effectively transformed the system's design into a fully functional, cross-platform application. By leveraging modern technologies—React and Flutter for the frontend, ASP.NET Core and Flask for the backend—the platform offers a responsive and user-friendly experience across devices.

Key modules, including AI-based baby cry analysis, the marketplace, chat, and guidance features, were successfully developed to address mothers' practical needs. The integration of machine learning via TensorFlow and secure data handling through SQL Server further ensured system intelligence, reliability, and scalability.

Overall, the implemented system meets its functional goals and provides a solid foundation for future development and deployment in real-world maternal care contexts.

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Appendix:

- This project has been submitted to the High Institute of Computer Science and Information Systems as part of the requirements for obtaining the Bachelor's degree in Computer Science and Information.
- The project aims to develop an integrated digital platform to support new mothers through a mobile application and a website. The platform offers a variety of smart services such as baby cry sound analysis using artificial intelligence, group chat functionality, an online marketplace for exchanging baby products, and daily refreshed tips related to child care.
- According to the instructions of the High Institute of Computer Science and Information Systems, the project has been uploaded to GitHub: https://github.com/FadyEzzat4923/Graduation_Project

ملخص المشروع

يهدف هذا المشروع إلى تقديم منصة رقمية متكاملة لدعم الأمهات الجدد في رحلة تربية أطفالهن الرضع، من خلال توفير مجموعة من الأدوات والخدمات الذكية التي تسهل عليهن فهم احتياجات الطفل اليومية والتفاعل معها بطريقة عملية وسريعة. تأتي المنصة في شكل تطبيق مخصص للهواتف المحمولة بالإضافة إلى موقع إلكتروني، مما يوفر سهولة في الوصول والاستخدام في أي وقت ومن أي مكان.

تُعنى المنصة بتوفير بيئة تفاعلية وآمنة تمكّن الأمهات من التواصل مع بعضهن البعض وتبادل الخبرات والتجارب، مما يعزز من الإحساس بالدعم المجتمعي ويقلل من مشاعر القلق والوحدة التي قد تواجهها الأمهات في المراحل الأولى من الأمومة.

يستطيع كل مستخدم إنشاء حساب شخصي خاص به، مما يتيح له الوصول إلى كافة خدمات المنصة والتفاعل مع باقي المستخدمين. وتوفر المنصة نظامًا خاصًا للمحادثات الجماعية، يتيح للأمهات التحدث معًا في غرف حوار مفتوحة لمشاركة الخبرات، طرح الأسئلة، وتقديم النصائح، مما يخلق مجتمعًا متعاونًا ومترابطًا.

كما تحتوي المنصة على قسم خاص لعرض وبيع مستلزمات الأطفال، حيث يمكن لكل أم رفع صور المنتجات التي ترغب في بيعها، مع إمكانية التواصل المباشر بين الأمهات باستخدام أرقام الهواتف، دون الحاجة لوسطاء أو خطوات معقدة. هذه الخدمة تساهم في تسهيل تبادل المستلزمات بين الأمهات، وتوفير عليهن الوقت والجهد والتكاليف.

ومن أبرز مميزات المنصة وجود نظام ذكي لتحليل صوت بكاء الطفل، حيث يمكن للأم تسجيل صوت بكاء طفلها، فيقوم النظام بتحليل الصوت باستخدام تقنيات متقدمة لتحديد السبب المحتمل للبكاء مثل الجوع أو التعب أو الألم، ثم يُعرض على الأم مجموعة من التوصيات العملية للتعامل مع الموقف بشكل سريع وفعال، مما يقلل من الحيرة ويعزز من استجابة الأم لاحتياجات طفلها.

وتضم المنصة أيضًا قسمًا خاصًا بالنصائح اليومية الموجهة للأمهات، حيث يتم عرض نصيحة مختلفة في كل مرة يتم فيها تحديث الصفحة، لتقديم محتوى متجدد ومتنوع يشمل مختلف جوانب رعاية الطفل مثل التغذية، النظافة الشخصية، النوم، التطور النفسي والعاطفي، وغيرها من الموضوعات التي تهتم كل أم.